GEK-65477

DRIVE SYSTEM INSTRUCTIONS

DIRECTO-MATIC® PLUS

DLC+ SEQUENCER MAINTENANCE MANUAL

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL@ ELECTRIC

DIRECTO-MATIC® PLUS
DLC + SEQUENCER MAINTENANCE MANUAL

GEK-65477

ISSUE DATE: October, 1981

The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, U.S.A.'

Directo-Matic ®is a registered trademark of General Electric Company.
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HOW TO USE THIS MANUAL

This manual is a guide to maintaining General Electric's Directo-Matic® Plus Distributed Logic Control (DLC+) Sequencer and Programming Module. We have included information on installation, theory of operation, maintenance, and troubleshooting.

We have written this manual for the industrial electrician or electronics technician who must install and maintain the DLC+ Sequencer hardware. We assume that you have some background in digital electronics and that you have read the ***Directo-Matic® Plus DLC+ Sequencer Operator Manual (GEK-73906).*** We strongly suggest that you read the entire maintenance manual before attempting to install, troubleshoot, or repair the DLC+ Sequencer.

This manual contains four chapters and several appendixes. Chapter 1 introduces you to the DLC+ Sequencer and Programming Module and provides equipment specifications. Chapter 2 contains block diagrams and a theory of operation for the sequencer and programming module. Chapter 3 contains information on equipment maintenance, including installation, maintenance features, diagnostics, and spare parts. Chapter 4 contains troubleshooting information for the sequencer and programming module. The appendixes contain supplementary information such as error codes, error messages, and printed wiring board descriptions.

The following General Electric publications are suggested as reference material:

- Directo-Matic® Plus DLC+ Sequencer Operator Manual (GEK-73906)
- Control Signal Highway Maintenance Manual (GEK-65411)
- Directo-Matic *Plus Highway Test Module User Manual (GEK-6515 1)
- **System** documentation (elementaries, specifications, and backsheets) included with your equipment

WARNING

This equipment contains a potential hazard of electricalshock or burn. Only personnel who are adequately trained and thoroughly familiar with the equipment and these instructions should install, operate, or maintain this equipment.

WARNING

To avoid potential electrical shock and misoperation, the DLC+ Programming Module must be properly grounded through the three-prongpowerplug. Do not use two-prong "cheaterplugs". Also ensure that the electrical outlet in use has been properly grounded.

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CHAPTER 1
INTRODUCTION

General Electric's DLC+ Sequencer is a microprocessor-based programmable controller that performs decision-making and arithmetic functions under software control. Associated with the DLC+ Sequencer is a wide range of input/output boards that interface the sequencer to industrial components such as switches, motors, solenoids, and relays. The sequencer is part of the Directo-Matic ® Plus family of integrated industrial control equipment.

The DLC+ Sequencer is supported by a programming module, a cassette tape drive, and a printer. These support devices are used to program and troubleshoot the DLC+ Sequencer and its I/O equipment. The programming module allows you to inspect and modify sequencer software, control the operation of the sequencer, and run sequencer diagnostics. The cassette tape drive is used to load application programs into the sequencer memory and dump the contents of sequencer memory to cassette tape for storage. The printer is used to produce copies of application software in relay ladder diagram form. You will find complete information on the use of these devices in the ***Directo-Matic® Plus DLC+ Sequencer Operator Manual (GEK-73906)***.

EQUIPMENT SUMMARY

As mentioned, the DLC+ Sequencer is a programmable controller. A programmable controller is a specialized computer used for sequence control of an industrial process. Programmable controllers are rugged, reliable, cost effective, and easy to program. A small programmable controller can perform the control and sequencing jobs that once required hundreds of relays.

Figure l-l is a simplified block diagram of a programmable controller and its supporting modules. The **memory** stores customized application software for the particular process being controlled. The **input interface** accepts inputs from process hardware such as limit switches, operator controls, and sensors and converts these inputs to logic signals for use by the processor. The **processor** scans these input signals and sends commands to the output interface under application software control. The **output interface** converts these commands to signals understood by devices such as motor starters, contactors, solenoids, and operator indicators. The **programming module** is used by an operator or programmer to monitor and modify the application program in memory or to troubleshoot a system problem. The **tape drive** is used to load application programs into memory or dump the contents of memory to cassette tape.

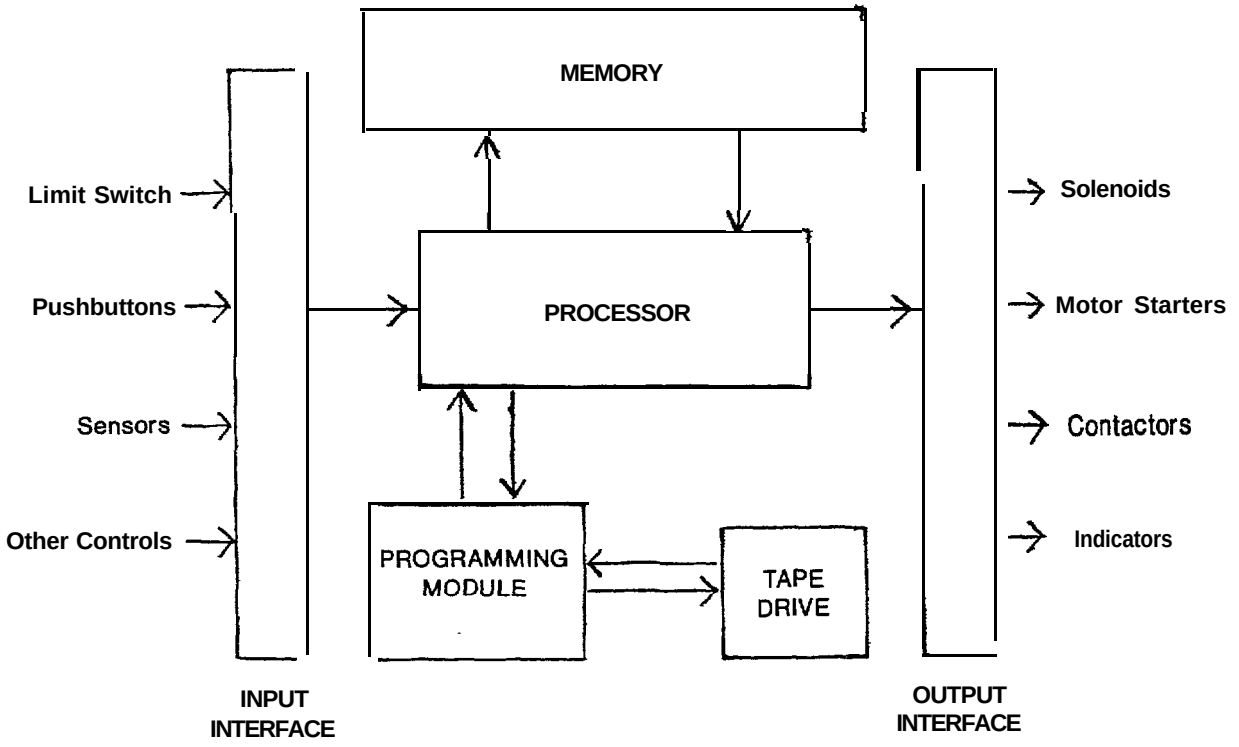


Figure l-l
Programmable Controller Simplified Block Diagram

The programmable controller constantly scans its inputs and updates its outputs. When the processor detects a change in an input, it uses application software instructions to determine what action to take. For instance, if a motor start button on an operator console (an input device) is pressed, the processor (under application software control) may check to see if all appropriate interlocks are closed. If they are all closed, the processor sends a signal *to* a motor control center (an output device) telling it to start the desired motor.

Since the process-controlling logic in a programmable controller is in application software, equipment additions or process changes are accomodated by simply changing the software, thus avoiding expensive rewiring. In the DLC + Sequencer, application software is easily modified by means of industry-standard relay ladder diagrams.

General Electric's DLC+ Sequencer and DLC+ Programming Module are part of a growing family of Directo-Matic® Plus control equipment. Some of the characteristics of these two DLC+ products are listed below.

- Both of these products are microprocessor-based, which means that product operation is controlled primarily by software or firmware rather than hardware.
- Automatic board-level diagnostics identify hardware faults on all microprocessor-based boards through a yellow IMOK light located on the front edge of the board. If the light goes out, a hardware fault has been detected.

- A wide range of I/O boards allows a variety of input and output devices to be connected to the sequencer, including digital and analog devices.
- The sequencer is capable of remote communications over time-shared wiring with I/O devices, other sequencers, and process control computers.
- The programming module is powerful, portable, and easy to use. Application programs are displayed in easy-to-understand relay ladder diagram form.

EQUIPMENT CONFIGURATIONS

A typical sequencer configuration contains standardized "building blocks" arranged to perform the desired process control functions. These "building blocks" include printed wiring boards (such as I/O, microprocessor, memory, and communication boards) and electronic modules (such as power supplies, card cages, and isolation modules). DLC+ Sequencer building blocks are described in Chapter 2 of this manual.

Although the DLC + Sequencer can be used in a variety of process control applications, this section will cover only two basic configurations: the DLC+ Sequencer as a stand-alone programmable controller and as part of a hierarchical process control system.

In its simplest configuration, the DLC+ Sequencer operates as a stand-alone programmable controller to coordinate local and remote I/O devices, drives, and operator desks. As shown in Figure 1-2, the sequencer can communicate with local I/O devices directly, and with remote I/O devices over a Control Party Line. The sequencer can also communicate directly with Siltron™ power conversion modules. Local or remote I/O boards may be mounted in a field termination module or an electronics module (these terms will be discussed in Chapter 2). The blocks in Figure l-2 marked LIMA represent isolation modules used for long distance communication over the Control Party Line (CPL).

The sequencer can also be tied into a hierarchical process control system as shown in Figure 1-3. In this configuration, the sequencer becomes a remote slave station under the master control of a process computer. Communications with the process computer take place over a Control Signal Highway. The sequencer still coordinates and controls all I/O devices attached to it, but does so *under the overall control of a process computer.

The above configurations are only two of the many possible DLC + Sequencer configurations available. The system elementary included with your equipment contains information regarding the particular configuration of your system. This maintenance manual contains information on the "building blocks" that make up your system. Together these two documents will help you understand, troubleshoot, and maintain your DLC+ Sequencer system.

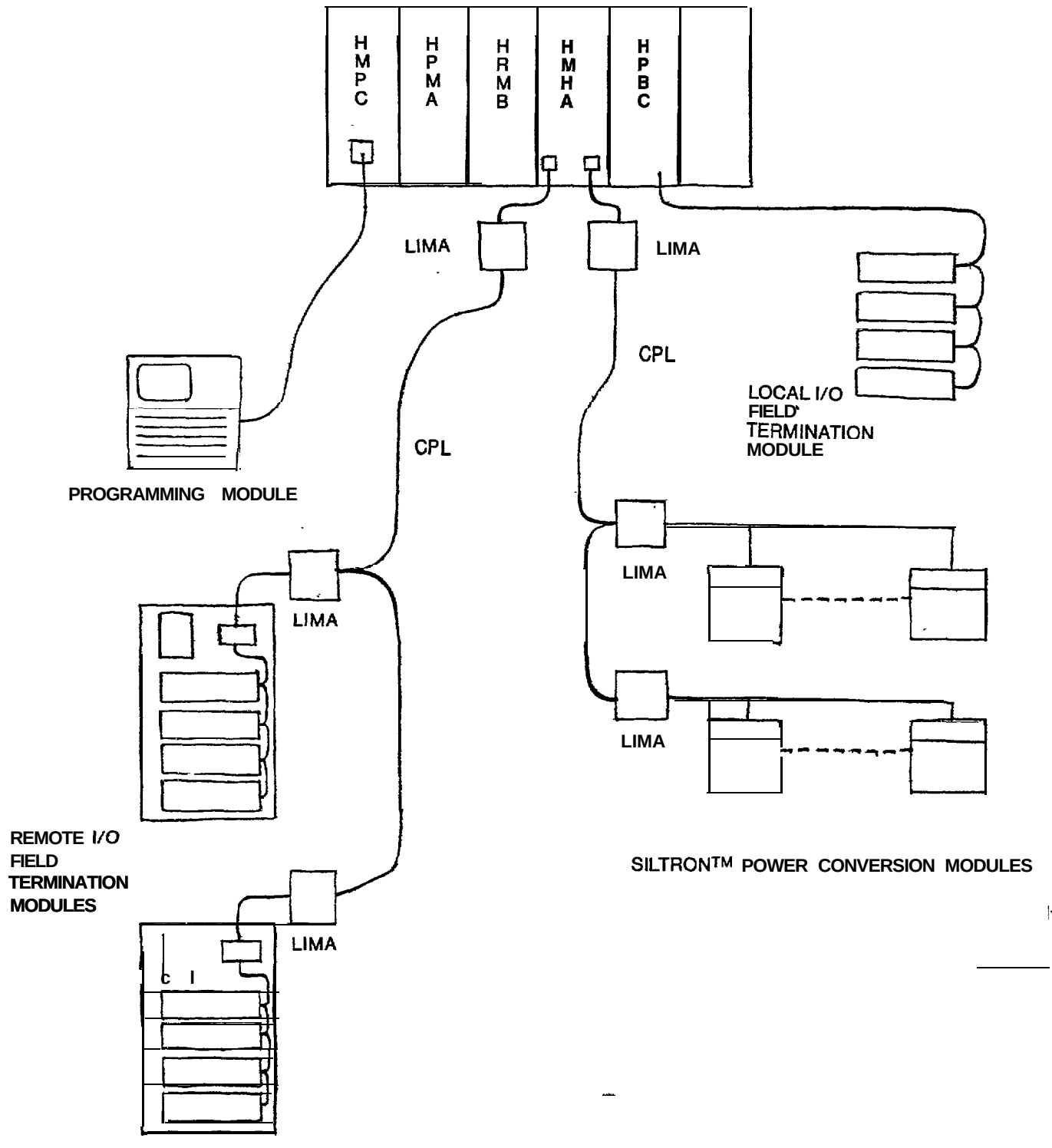


Figure 1-2.
DLC + Sequencer Configured as
Stand-Alone Programmable Controller

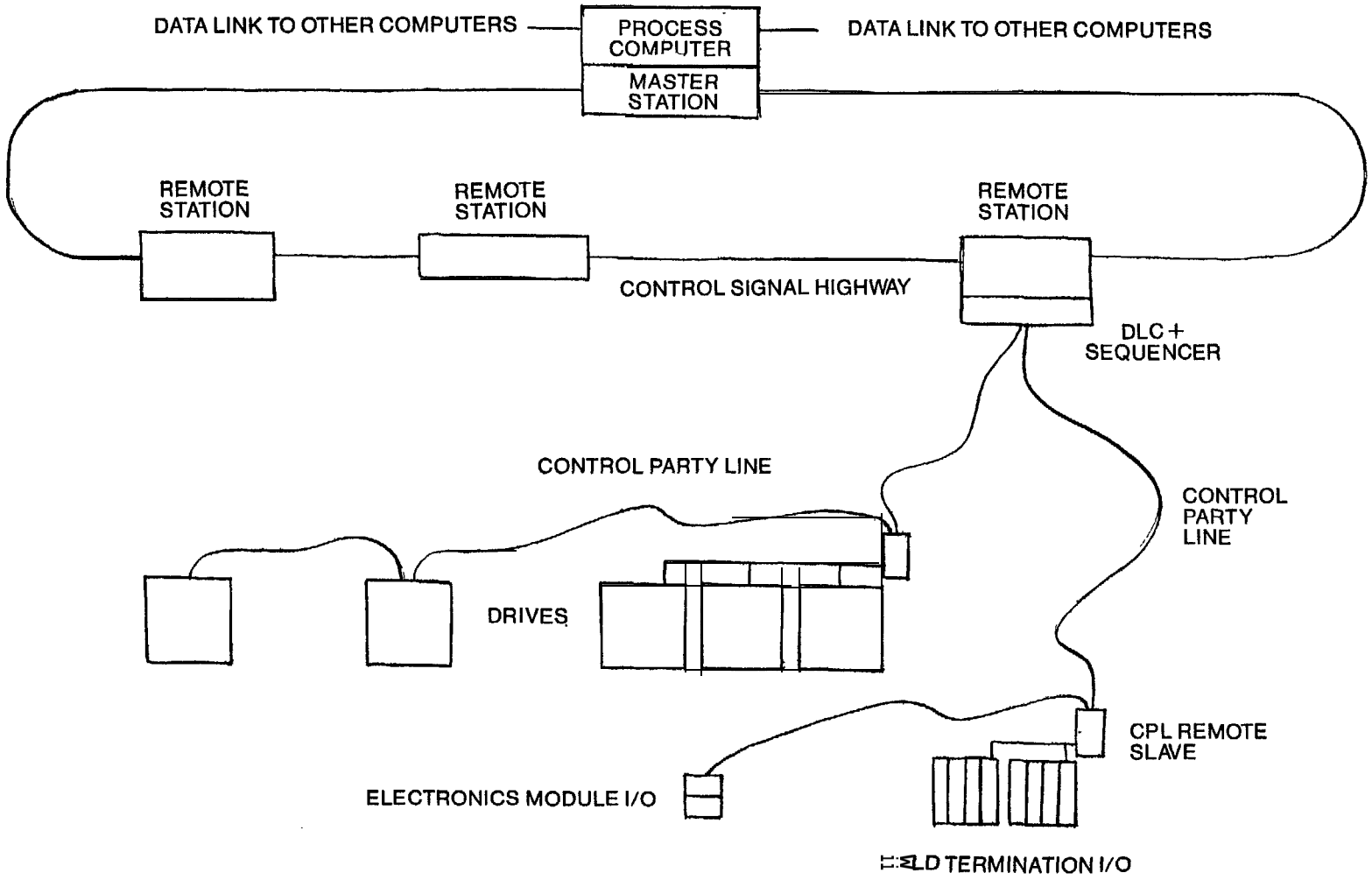


Figure 1-3.
DLC+ Sequencer in Hierarchical Process Control System

EQUIPMENT SPECIFICATIONS

The equipment specifications listed below are of a general nature. For more specific information, refer to the system specification supplied with the equipment.

INPUT POWER

Sequencer and I/O Modules (through PS5 Power Supply): single phase 87 to 127 VAC, 47 to 63 HZ)
Programming Module: 115 VAC (±10%), 60 HZ (± 5%)
Tape Drive and Printer: see operation manual for those devices

COOLING

The DLC + Sequencer and I/O Modules are convection-cooled. The DLC + Programming Module contains a built-in muffin fan for cooling. Proper ambient temperature must be maintained for effective cooling (see below).

OPERATING ENVIRONMENT

Ambient Temperature: 0°C to +40°C (+ 32°F to + 104°F)
Temperature Transients: less than ± 15°C per hour (±27°F per hour)
Relative Humidity: 5% to 95% (noncondensing)
Altitude: sea level to 1066 meters (3500 feet)
Atmosphere: Normal atmosphere found in typical heavy industry control equipment locations.

STORAGE ENVIRONMENT

Ambient Temperature: -20°C to +70°C (-4°F to + 158°F)
Relative Humidity: 0% to 95% (noncondensing)
Altitude: sea level to 1066 meters (3500 feet)
Atmosphere: Normal atmosphere found in typical heavy industry control equipment locations.

RADIO FREQUENCY INTERFERENCE

This equipment is designed to operate in a normal industrial atmosphere. Operating radio transmitters nearby may cause the control equipment to misoperate. It is recommended that radio transmitters not be operated within the vicinity of the control equipment when the panel doors are open, nor at any time within the confines of a power control room.

CHAPTER 2
EQUIPMENT DESCRIPTION

This chapter provides a high-level description of the DLC+ Sequencer and DLC+ Programming Module hardware. The theory of operation of the sequencer and programming module boards is described in the applicable board backsheets and will not be covered here. As you read this chapter it may be necessary to refer to Appendix A, which contains a list and brief description of printed wiring boards used in the DLC+ Sequencer.

SEQUENCER EQUIPMENT

The DLC+ Sequencer is a highly structured programmable controller consisting of a number of hardware and software "building blocks." As additional functions or capabilities are needed, the sequencer is expanded by adding extra boards, modules, firmware (in PROM), and software (in RAM or PROM). Because of its modularity, the sequencer can handle a wide range of process control and sequencing needs.

The DLC+ Sequencer hardware can theoretically handle from 32 to 2000 I/O points, depending on the size of memory and the number of I/O boards installed. An I/O point is equivalent to a single I/O signal such as a limit switch input or an output to a solenoid. The sequencer can also handle groups of I/O points configured as BCD or integer values.

Internally, the sequencer can store and manipulate from 600 to 3600 elements within a number of relay strings. An element consists of a contact or coil plus its associated signal address. The maximum number of elements available in a particular installation depends on the size of sequencer memory.

The application software within the sequencer which controls the external process can reside in either RAM or PROM memory RAM-based application software can be modified by the programming module, while PROM-based software is permanent and cannot be changed. Application software for a process control system that is expected to change (by the addition or deletion of equipment, changing parameters, etc.) will typically be stored in RAM memory. When changes are no longer expected, the software can be programmed into PROM chips.

The sequencer processes I/O signals through a wide variety of I/O boards. Appendix A contains a list of the common I/O boards used in the sequencer system. These I/O boards can handle logic level (+5VDC) signals, high level AC and DC signals, and analog to digital and digital to analog conversion. I/O devices can be located up to 4000 feet from the sequencer module.

SEQUENCER BOARDS

A variety of printed wiring boards make up the hardware building blocks used in the DLCS Sequencer. The boards are plugged into an electronics module (see "Sequencer Packaging" below) and require a power supply for operation. The physical location of each type of board in the electronics module is requisition-dependent and is covered in the system elementary sent with the equipment.

A minimum sequencer requires the following boards:

- HMPC Microprocessor Board - the "heart" of the DLC+ Sequencer
- HPMA 32K/64K PROM Memory Board(s) - stores product or application software
- HRMB 8K RAM Memory Board(s) - stores application software
- One or more I/O boards to interface I/O devices with the sequencer

The HMPC board contains an Intel 8086 microprocessor, which processes both application and product software during operation, In addition, the board contains RAM and PROM memory, plus I/O circuits for communication with other HMPC boards or the DLC + Programming Module.

The product software (DLC+ Sequencer operating system) is supplied in PROM memory (on an HPMA board). In most cases, the application program resides in battery-backed RAM memory for ease in programming changes during field installation and equipment modification. The application program may, however, be programmed into PROM memory (on HPMA boards) at the factory if the program is not expected to change in the future. The DLC+ Sequencer must contain at least one HPMA board (containing product software) and one HRMB board; it may contain up to five extra HRMB boards or one extra HPMA board (containing application software).

The I/O boards for a very small system would probably be located in the same electronics module where the microprocessor and memory boards are. Additional hardware allows the I/O boards to be located outside the module and even thousands of feet away. See the section below titled "Sequencer I/O Structure" for more information. Further information on a specific board can be found in the backsheets issued for that board.

, When required, the DLC + Sequencer can communicate with a high-level process control computer over a Control Signal Highway. Additional boards are required to convert the sequencer into a CSH slave station.

SEQUENCER BUS STRUCTURE

The DLC+ Sequencer hardware is designed around four buses:

- **L-bus (Local bus)** - The main sequencer bus through which the microprocessor (HMPC) board communicates with boards located within a single electronics module. These boards might include memory (HPMA and HRMB) boards, Control Party Line master (HMHA) boards, and module I/O boards. The L-bus is located within the backplane of the electronics module.
- **E-bus (Express bus)** - A data bus used for communications between two or more DLC+ Sequencer HMPC boards, or between an HMPC board and a Control Signal Highway (CSH) slave interface.
- **F-bus (Full bus)** - A sequencer I/O bus containing address, data, and control lines.
- **R-bus (Reduced bus)** - A partial decode of the F-bus used for communication with I/O boards via the Control Party Line (CPL). The HSHA CPL slave can provide either the F-bus or R-bus to I/O boards in a particular electronics module.

The L-bus, E-bus, F-bus, and R-bus are illustrated in Figure 2-1. In this figure, a DLC+ Sequencer (HMPC #1) is connected to another DLC + Sequencer (HMPC#2) via daughter boards (DMPC) and the E-bus. All communication between the two sequencers occurs over this link. Note that the two HMPC boards use separate L-buses to communicate with their associated memory and I/O boards. A Control Party Line is implemented by an HMHA master board connected to an HSHA slave board. The HSHA board could develop an F-bus which is converted to an R-bus by the HPBC board, or the HSHA could develop the R-bus directly which is then used by the I/O boards within the CPL slave's card cage.

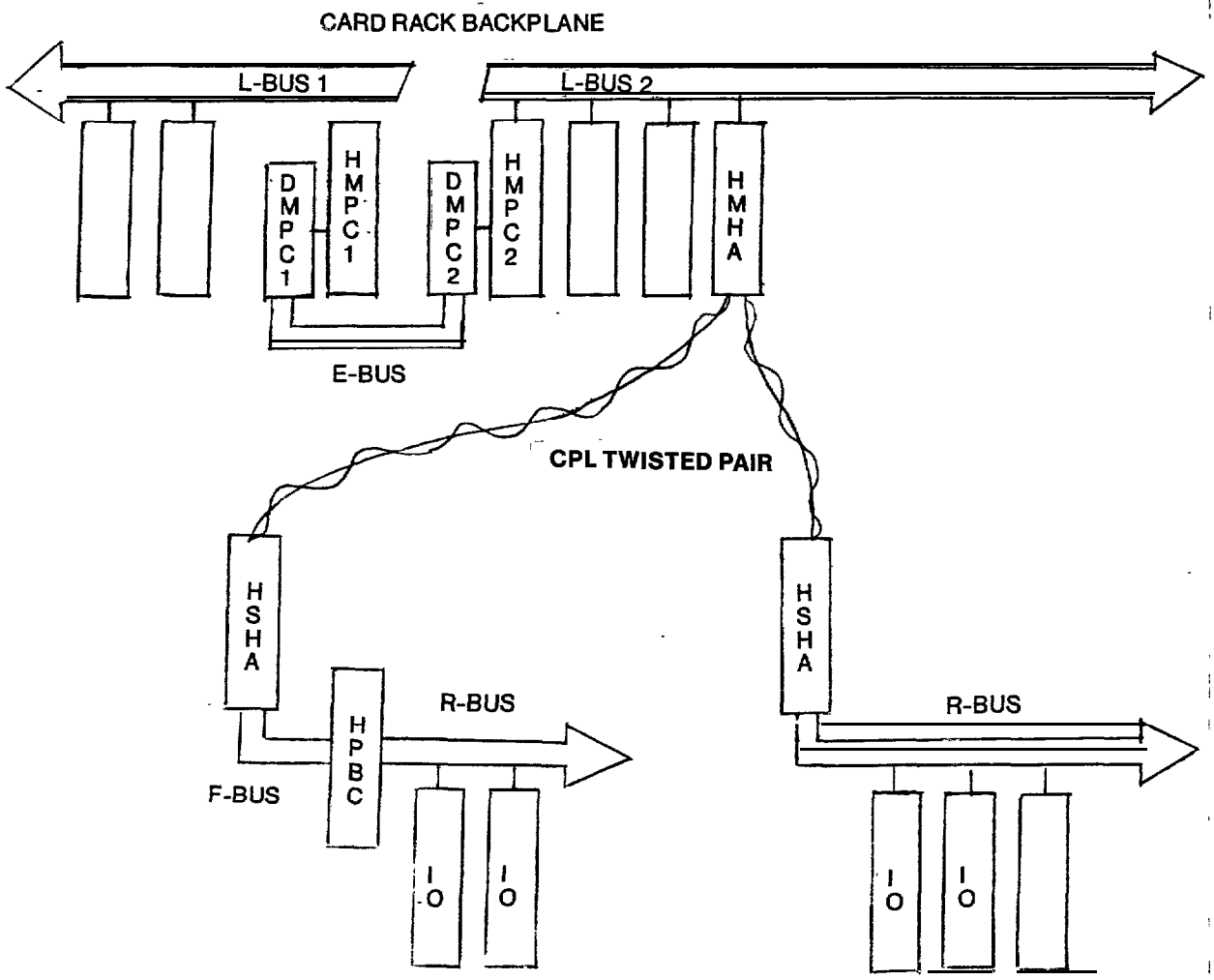


Figure 2-l.
Diagram of DLC + Sequencer Bus Structure

SEQUENCER I/O STRUCTURE

The DLC+ Sequencer handles three different types of I/O, classified according to the distance between the I/O board and the sequencer module:

1. Module I/O - I/O boards located in the same module as the sequencer
2. Local I/O - I/O boards located in a separate module within 50 feet of the sequencer module
3. Remote I/O - I/O boards located in a separate module more than 50 feet from the sequencer module

When module I/O is used, the I/O boards are inserted in the same card rack as the sequencer boards, as shown in Figure 2-2. The I/O boards are interrogated directly by the HMPC and no buffering is required. Addressing is handled automatically by backplane wiring.

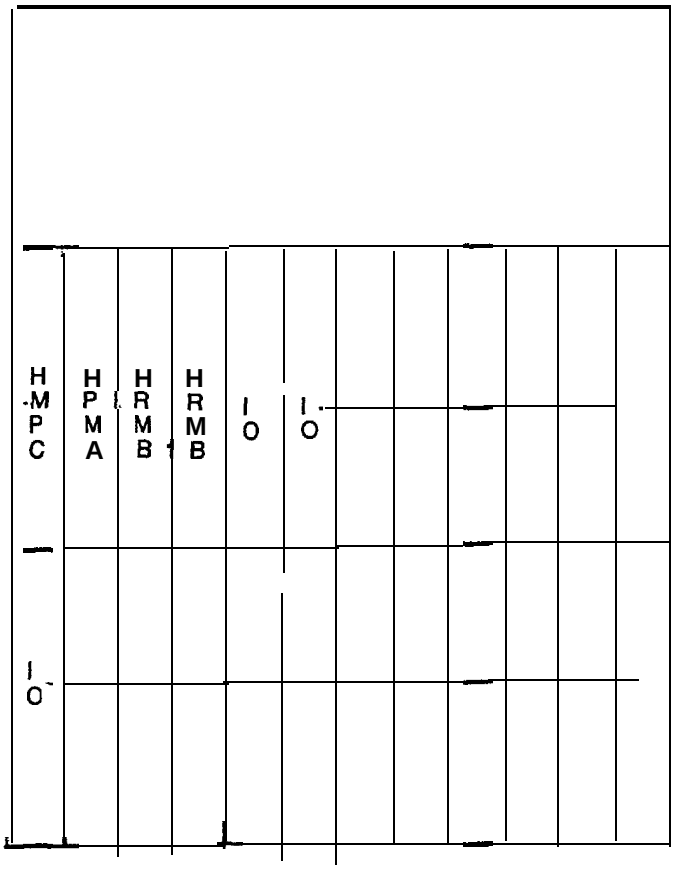


Figure 2-2.
Module I/O Configuration

When local I/O is used, the I/O boards are inserted into a separate card rack from the sequencer boards, as shown in Figure 2-3. This card rack can be an electronics module or a field termination module and can be located up to 50 feet from the sequencer module.

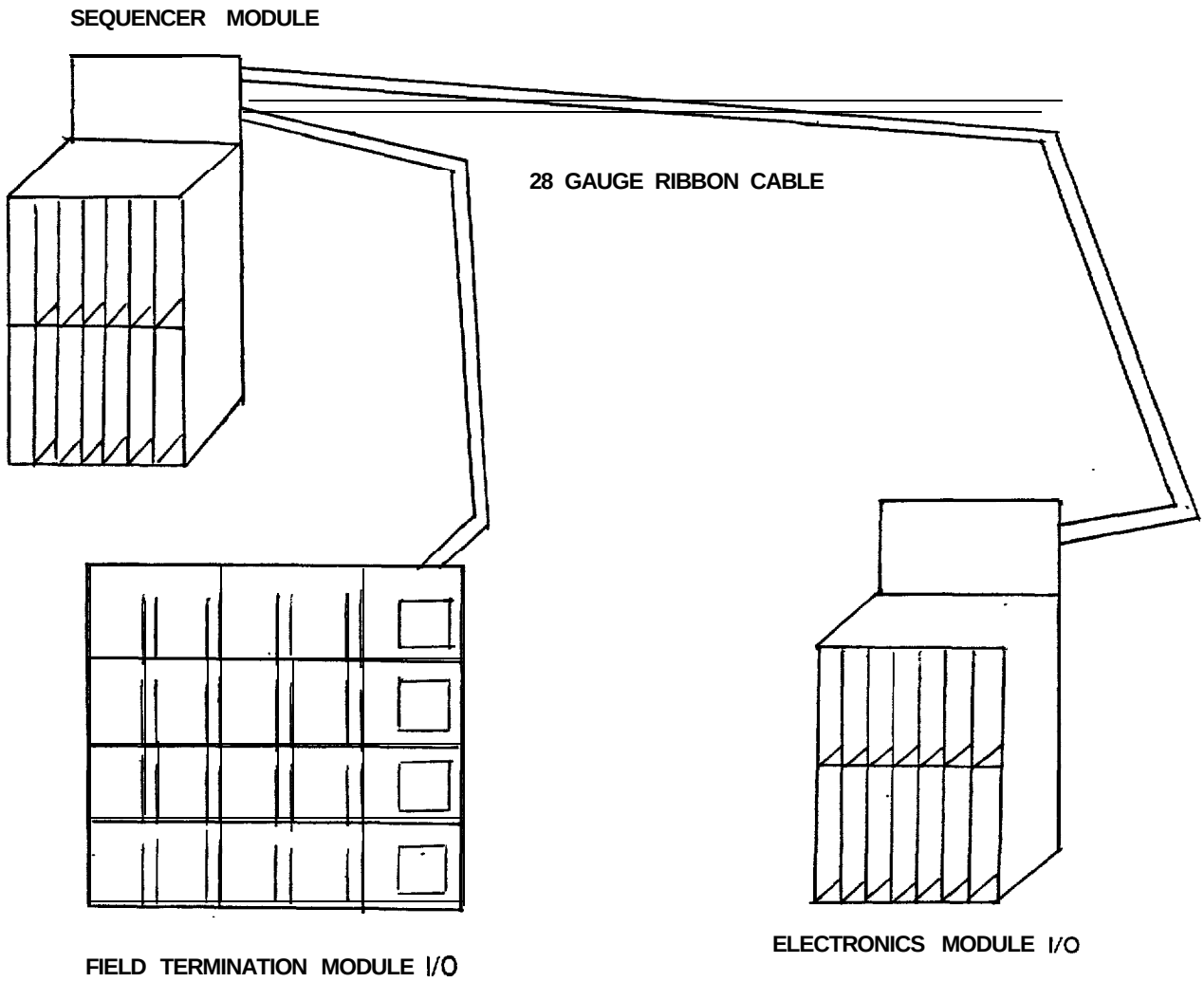


Figure 2-3.
Local I/O Configuration

Remote I/O communication is accomplished by incorporating one or more Control Party Line (CPL) communication links into the sequencer system as shown in Figure 2-4. A CPL is implemented by an HMHA master board in the sequencer module connected to one or more more I/O modules (using an HSHA board as a slave interface) or Siltron™ power conversion modules (using an HIOB board as a slave interface) by two twisted-pair shielded cables. Theoretically, up to 64 modules can be handled by one HMHA board, although the practical limit is much less. The modules can be located up to 4000 feet from the sequencer module. If remote I/O stations are over 300 feet from the sequencer module, the CPL channels are optically coupled via LIMA modules to improve signal isolation and noise immunity. Up to four HMHA boards, each capable of controlling two Control Party Lines can be handled by **one** sequencer. Each HMHA allows full duplex, synchronous, serial communication over two CPL channels at a 48 Khz. rate on each channel. Data transmitted over the Control Party Line is automatically checked for validity and is rejected if an error occurs.

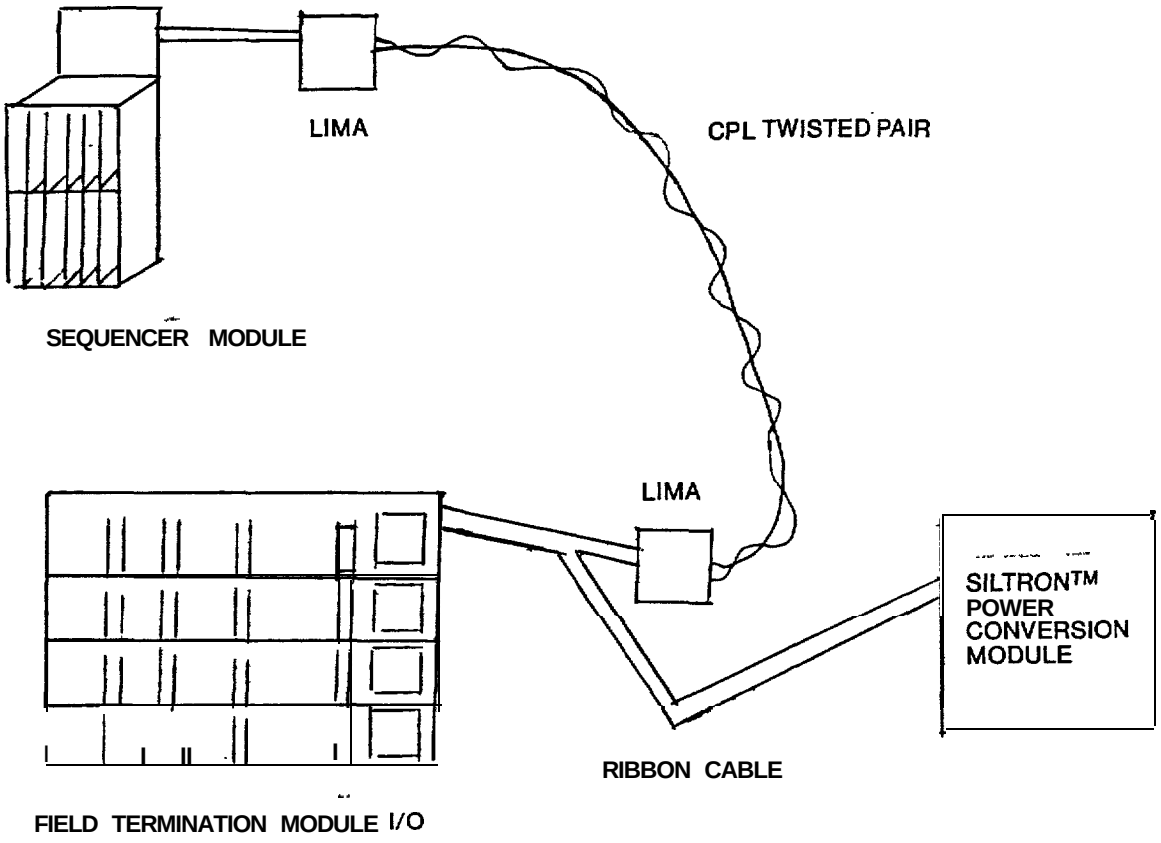


Figure 2-4.
Remote I/O Configuration

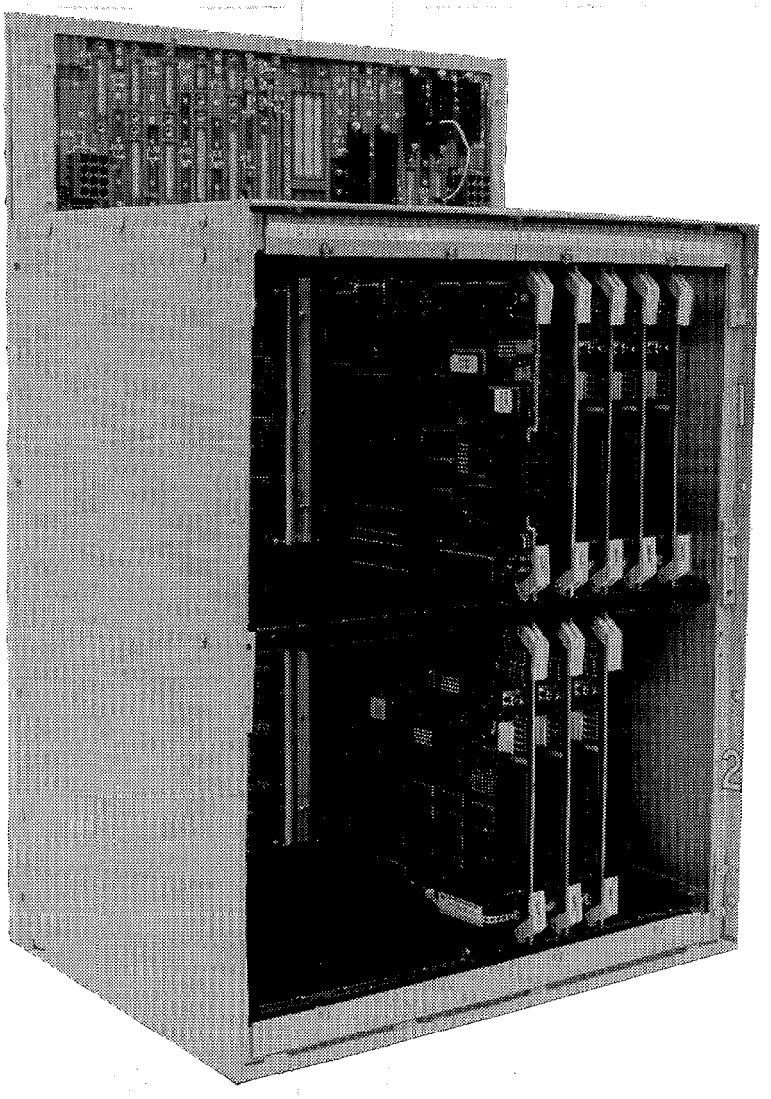
SEQUENCER PACKAGING

Two types of card racks are available for mounting DLC + Sequencer and I/O boards: the electronics module and the field termination module. The electronics module can contain either sequencer or I/O boards, while the field termination module contains only I/O boards.

A typical electronics module is shown in figure 2-5. There are three different sizes of electronics modules available:

- TEM (Tiny Electronics Module) - holds up to 6 boards
- SEM (Small Electronics Module) - holds up to 12 boards
- MEM (Medium Electronics Module) holds up to 24 boards

When an I/O board is installed in an electronics module, the wiring to the board from external devices is terminated with a 17 pin connector which plugs onto the front of the board. I/O boards in operator stations are typically installed in electronics modules rather than field termination modules due to the higher density of I/O signals in the operator station.

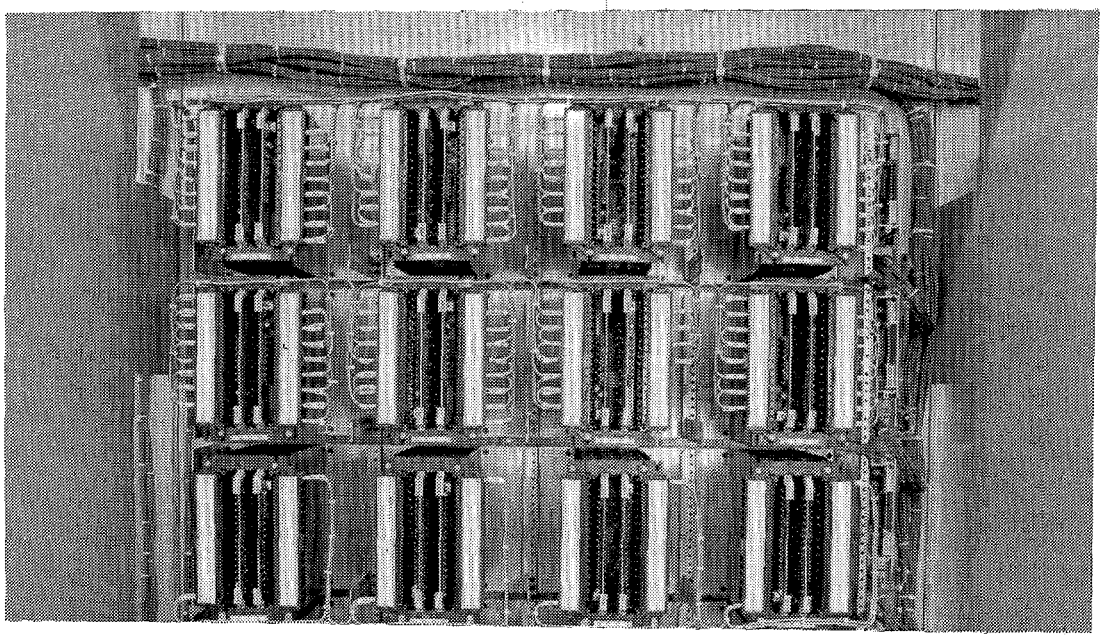


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Figure 2-5.
Typical Electronics Module

Figure 2-6 shows a typical configuration of field termination modules, which can be used for either local or remote I/O. Each module holds 4 or 8 I/O boards and up to 4 modules can be ganged together to hold up to 30 I/O boards (2 slots are reserved for interface boards). This arrangement is ideal for field termination of signal wires, since wire troughs and terminal strips for terminating 12 gauge wire are also supplied. I/O boards may be inserted and removed without disturbing the field wiring and terminations.

The field termination module accepts digital I/O boards; analog I/O boards can be used with some restrictions. Digital or analog I/O boards can be used without restrictions in an electronics module.



(DSF-54256)

Figure 2-6.
Typical Field Termination Module

POWER SUPPLIES

General Electric provides most power supplies required by sequencer components. DC digital input power is provided from -105VDC unregulated power supplies in current ratings of 2 and 4 Amperes. AC digital output power (for use with solenoids and other such devices) is available from an AC source in the motor control centers. DC power required by board electronics is provided by a PS5 power supply (described later in this chapter) that provides +5VDC at 40 Amperes, ± 15VDC at 2 Amperes, and +28VDC at 5 Amperes. Other power supply ratings are available for special applications. All power supplies require a single-phase, 115 VAC, 50/60 Hz. source.

SEQUENCER DIAGNOSTIC CONCEPTS

The DLC+ Sequencer was designed with system uptime as a prime consideration. The components used in the sequencer were selected with reliability in mind and the total number of system components was minimized to reduce the probability of failure. However, since any electronic system will occasionally fail, powerful diagnostic and maintenance aids were designed into the DLC+ Sequencer to enable quick location and replacement of faulty hardware. Some of these diagnostic aids are listed below.

- The sequencer and programming module perform self-tests on power up and can report faults via the programming module screen.
- Each board containing a microprocessor (including the HMPC, HMHA, and HSHA boards) is equipped with a yellow IMOK light. As long as the board operates properly, the IMOK light remains on, but if a failure occurs, the IMOK light goes off. This makes troubleshooting many problems simple.
- All I/O boards contain a status indicating light for each input or output signal. An operator or maintenance man can tell at a glance whether a specific signal line is energized or deenergized.
- A DSHA daughter board containing maintenance aids is available for the HSHA CPL slave board. Using this daughter board, an operator or maintenance man can monitor or exercise I/O boards that are controlled by the HSHA slave board.
- The DLC+ Sequencer Programming Module allows on-line monitoring of inputs and outputs, the ability to force outputs, and the ability to run sequencer tests.
- A Highway Test Module is available for testing and troubleshooting the Control Signal Highway.

Chapter 3 of this manual provides more information on these diagnostic and maintenance aids.

SEQUENCER DESCRIPTION OF OPERATION

When power is first applied, the sequencer executes a system self-test (described in Chapter 3) that checks the operation of most system components. After the self-test successfully executes, the sequencer system either waits for application software to be loaded from tape (on a RAM-based system) or immediately begins to execute application software (on a ROM-based system).

During application software execution, several actions occur in sequence, controlled by both application and product software. The following sequence of events assumes that the sequencer contains at least one Control Party Line (CPL) containing an HMHA and at least one HSHA with associated I/O boards.

1. The HMPC board sends a command to the HMHA board asking it to "poll" or interrogate its slave stations (HSHAs). During this poll, the HMHA accepts input data from the slave stations and provides output data to the slave stations; since the CPL is full duplex, transmission of input and output data can occur simultaneously. The slave station in turn accepts inputs from the I/O boards attached to it and provides outputs to those same I/O boards. Thus, an input signal from a specific I/O board on a Control Party Line must go through an HSHA board, over a twisted pair wire, through an HMHA board, through the L-bus on the sequencer backplane, and finally on to the application program within the sequencer CPU (i.e., the HMPC and its memory). Likewise an output signal originating in an application program within the sequencer CPU must travel through the L-bus, through the HMHA board, over a twisted pair wire, through an HSHA board, and on to the particular I/O board. Each input and output signal represents a software contact or coil within the application software. The sequencer handles addressing of the correct input or output board by means of a signal identification scheme described in the ***Directo-Matic® Plus DLC+ Sequencer Operator Manual (GEK-73906). This*** entire operation is known as an "I/O sweep".

2. Once the sequencer CPU has received inputs from all I/O boards (whether module I/O, local I/O, or remote I/O), it updates the status of software contacts within the application program. That is, if a normally-open contact was originally open, but its associated input signal became energized, the contact would close.

3. Once all contacts have been updated, the sequencer CPU "solves" all of the relay logic within the application software, resulting in updated coil (output) signal status. For instance, a newly-energized contact might cause an output coil to pick up. This information must then be transmitted to the associated I/O board.

4. At this point, the sequence of events starts over again. Input data is accepted from I/O boards and output data is sent to I/O boards. In reality, the HMHA and HSHA boards are constantly monitoring their I/O boards in parallel with sequencer operation, not only at specific times.

5. If the sequencer is tied into a Control Signal Highway (CSH) link, all of the above must occur along with communications between the sequencer and the CSH link. For more information on the CSH, refer to the ***Control Signal Highway Maintenance Manual (GEK-65411)***.

PROGRAMMING MODULE EQUIPMENT

As shown in Figure 2-7, the DLC+ Programming Module contains standard DLC+ Sequencer boards along with a power supply, video monitor, and keyboard. The HMPC microprocessor board coordinates the operation of the programming module under control of product software stored in PROMS on the HPMA PROM memory board. The HMPC also provides an interface for communication with the DLC+ Sequencer. The HRMB RAM memory board stores strings being created and also acts as a communications buffer for the programming module. The HVDA video display board acts *as* an interface for the video monitor and keyboard. In addition, the HVDA interfaces a tape drive or printer and provides a video output jack for an external video monitor. The power supply provides necessary AC and DC voltages for proper operation of the programming module.

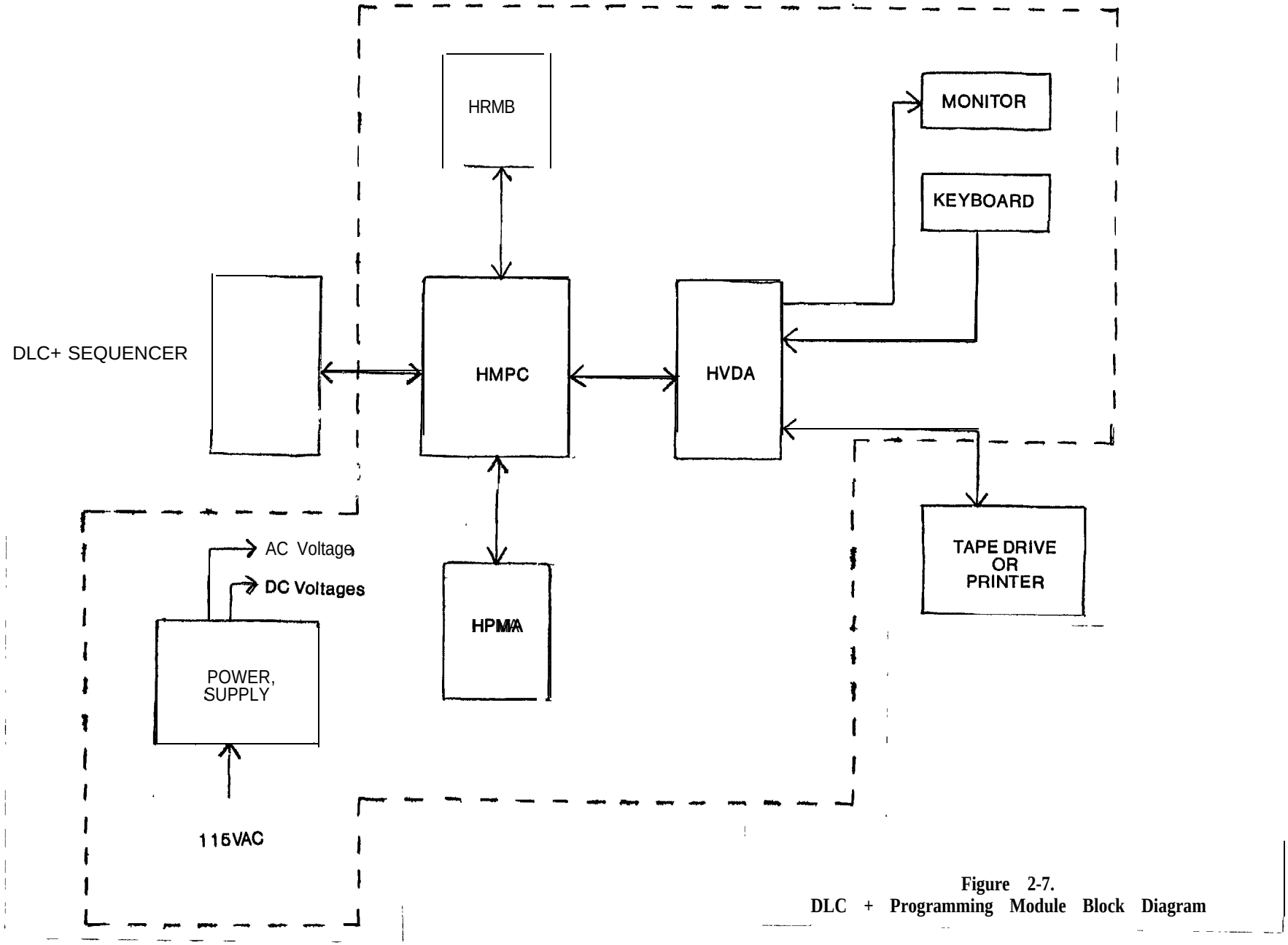


Figure 2-7.
DLC + Programming Module Block Diagram

MODULE DESCRIPTIONS

This section of the maintenance manual contains brief descriptions of the modules used in conjunction with the DLC+ Sequencer. For more information on these modules, see the detailed module backsheets shipped with your equipment.

ELECTRONICS MODULE

Several types of electronics modules are used in the DLC+ Sequencer system. Some types of electronics modules (the DS3800SDS, for example) hold sequencer boards (microprocessor, memory, and communication boards), while some types hold I/O boards only. The electronics modules used in your particular system will be identified on the system elementary diagram and described in greater detail in the module backsheets.

There are three different sizes of electronics modules available: the TEM (Tiny Electronics Module), which holds up to 6 boards; the SEM (Small Electronics Module), which holds up to 12 boards, and the MEM (Medium Electronics Module), which holds up to 24 boards. Figure 2-5 shows a typical MEM module.

All electronics modules requires DC power from a PS5 power supply module.

FIELD TERMINATION I/O (HIO) MODULE

The field termination input/output (HIO) module is a horizontal assembly containing card slots for up to eight I/O boards. Up to four HIO modules can be panel-mounted together. Three HIO modules are shown in Figure 2-6. Each HIO module contains an HACB address customizing board in a card box at the end of the module. The HACB board is a wire-wrap board used to assign I/O addresses to each of the eight card slots in the module. A connector at the end of the HIO module connects the module into the DLC+ Sequencer system.

The HIO module has two main forms, one for local I/O and one for remote I/O (Control Party Line). The local I/O version can contain I/O boards in all eight slots. The remote I/O version contains an HSHA CPL slave board in slot 2, slot 1 must be empty, and the other six slots can contain I/O boards.

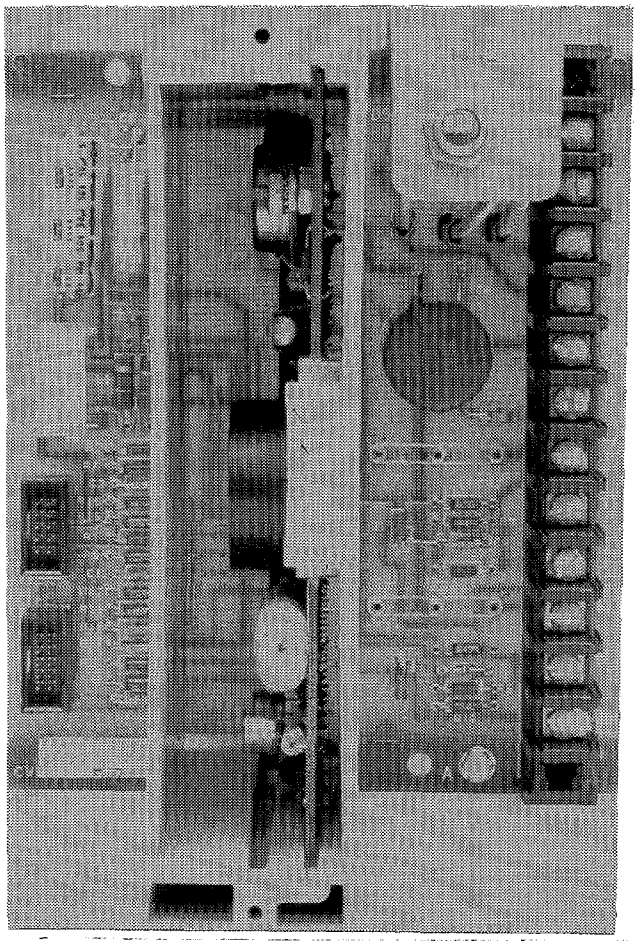
I/O boards handling 4, 8, or 16 input or output lines can be handled by the HIO module, but the module itself cannot handle more than 64 input or output lines (this is equivalent to eight boards each containing 8 input or output lines). Each of the available slots can contain either an input or an output board.

The HIO module requires DC power from a PS5 power supply module.

ISOLATION (LIMA) MODULE

The isolation (LIMA) module, shown in Figure 2-8, is an optical isolation device used to isolate and terminate signals on the Control Party Line (CPL) synchronous communication link. The LIMA module is typically used on long-distance CPL cable runs of 300 feet or more to provide common mode isolation and line suppression. The module provides full duplex communication through a single channel at up to 50 Kilobaud (for runs of 4000 feet or less). A standard telephone jack is provided on the module and may be used for voice communication between modules when connected to a suitable cable.

The LIMA module uses + 15 VDC and -15 VDC from the PS5 power supply to develop a +5 VDC regulated voltage that is used to power the module. A yellow IMOK LED monitors the status of the +5 VDC output; as long as the + 5 volts is present, the LED remains lighted.

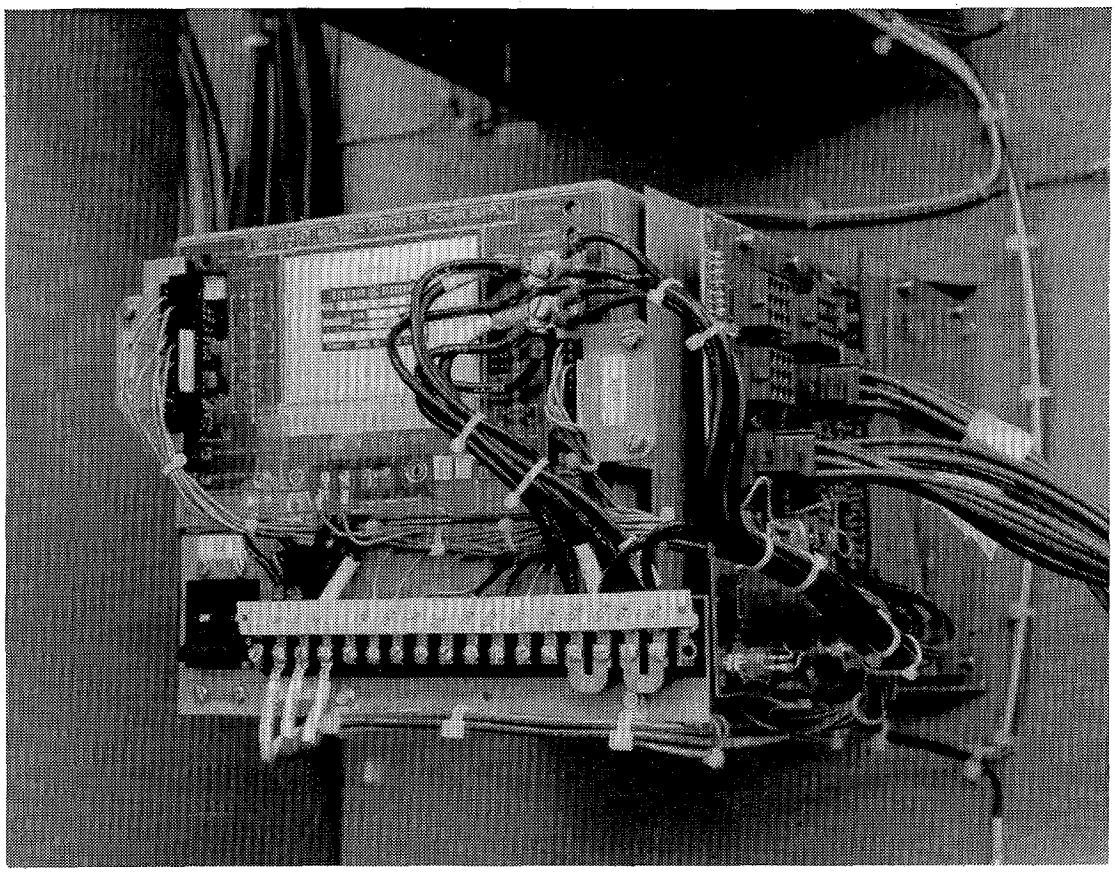


(DSF-54171)

Figure 2-8.
LIMA Isolation Module

POWER SUPPLY (PS5) MODULE

The power supply (PS5) module supplies DC power to electronics modules, field termination modules, and isolation modules in the DLC+ Sequencer system. The module, which is shown in Figure 2-9, contains a switching regulator along with control, monitor, and protection circuits. The PS5 supply accepts single phase AC voltage (87 to 127 VAC, 46-63 HZ) and provides four regulated DC voltages (+5 VDC, +28 VDC, + 15 VDC, and -15 VDC) at the six output jacks JA through JF. A "Ready" light on the power supply indicates that all output voltages are at their proper levels. Should any of the output voltage levels change significantly, the "Ready" light will go out and the power supply will shut down. During this shutdown procedure, a status signal (LVSL) will indicate the pending shutdown to those modules connected to the supply. After a short delay (.3 MS to 1.3 MS) to allow an orderly shutdown procedure in each module, all output voltages will be crowbarred simultaneously (the circuit breaker will not open). A loss of AC power will cause signal LVSL to indicate a pending shutdown before any DC voltage is affected. The PS5 supply can be reset after a shutdown by turning the circuit breaker off for 3 seconds and then back on.



(DSF-54255)

Figure 2-9.
PS5 Power Supply Module

CHAPTER 3
EQUIPMENT MAINTENANCE

This chapter contains maintenance information for the DLC+ Sequencer and Programming Module, including a description of maintenance features, diagnostics, and procedures for receiving and storing the equipment.

RECEIVING, STORING, AND INSTALLING THE EQUIPMENT

DLC+ Sequencer equipment is usually mounted in one or more free standing panels. You will probably receive the equipment in a packing case or crate. The packing case is not suitable for storing the equipment outdoors. If the equipment is not to be installed immediately, store it under cover in a clean, dry location away from areas where construction is in progress. Ensure that the storage environment meets the specifications listed under "Equipment Specifications." If the equipment is not to be installed immediately, store it under cover in a clean, dry location away from areas where construction is in progress. Ensure that the storage environment meets the specifications listed under "Equipment Specifications" in Chapter 1. If the equipment must be stored in an unusually damp area, cover the equipment and provide a heat source to prevent condensation. Should the equipment be exposed to moisture, it should be checked by General Electric service personnel before being put into service.

Examine each shipment carefully upon arrival and report any shortage or damage to the carrier and to the nearest General Electric sales office.

The equipment should be placed as close as possible to its permanent location before unpacking. Unpacking should be done with care to prevent damage or loss of parts. Be especially careful when lifting or moving the equipment. Refer to the equipment outline drawings for preferred methods of lifting and moving.

Before any installation work is done, consult all drawings furnished by General Electric as well as all applicable contract drawings for your particular installation. Pay particular attention to the outline drawings as you plan, install, or modify conduits and busways.

Proper grounding of the equipment is essential for safe operation. Connect each panel to a suitable earth ground, such as the structural steel support of your building. Ensure a good mechanical and electrical connection at the panel and the ground source.

WARNING

To prevent hazard of electrical shock or burn, approved grounding practices and procedures mast be adhered to.

INITIAL SET-UP AND CHECK-OUT OF EQUIPMENT

After the equipment has been physically installed, it should be set up and checked out as follows:

1. **With power off,** ensure that all signal and power cables are properly installed according to the system elementary and that all printed wiring boards are securely inserted into their respective backplanes.

2. Apply power and turn on all circuit breakers in the panel containing the sequencer module and in any local or remote panels containing I/O modules. All IMOK lights on the applicable printed wiring boards (HMPC, HMHA, HSHA, and HCMA) and on any LIMA isolation modules should come on within a few seconds. If an IMOK light fails to come on, refer to Chapter 4 of this manual for troubleshooting procedures.

3. Connect a DLC+ Programming Module to the DLC+ Sequencer electronics module as described *in* the ***Directo-Matic® Plus DLC+ Sequencer Operator Manual (GEK-73906)***. Power up the programming module and place the module in "Build String" mode. Refer to Chapter 4 if any error messages appear or if the programming module does not operate properly.

4. Unless the sequencer uses ROM-based application software, it will be necessary to load the application software into sequencer memory. Take the sequencer off-line, connect a cassette tape drive to the programming module, and load the applicable software from tape (as described in GEK-73906). Refer to Chapter 4 if the cassette tape drive does not operate properly or if the software fails to load.

5. Verify that the software loaded correctly by performing a tape/memory compare or by visually checking a sufficient number of strings using the programming module. Place the sequencer on-line. The process should now be under control of the application software. If any problems develop, refer to Chapter 4 for troubleshooting procedures.

PREVENTIVE MAINTENANCE

The DLC+ Sequencer and Programming Module require no preventive maintenance other than an occasional cleaning of the programming module viewing screen.

SPARE PARTS

General Electric suggests that you maintain a stock of spare parts for your DLC+ Sequencer and Programming Module. Stocking spare parts will ensure rapid replacement of malfunctioning parts and reduce potential downtime. The Renewal Parts Quotation Booklet, which was part of the documentation shipped with your equipment, contains a list of suggested spare parts.

There are two classes of spare parts: start-up spares and running spares. Start-up spares are critical parts that should be available during installation and tune-up. Running spares should be kept on hand to replace parts that may malfunction during normal operation.

After you have used a spare part, it is important to replace that part in your stock of spares immediately. This may be done in one of three ways:

- During the warranty period, replace defective parts with General Electric warranty parts.
- After the warranty period, use the General Electric Repair-and-Return service.

- After the warranty period, many parts can be replaced through General Electric's Element Exchange Service. Parts covered by the Element Exchange Service include most printed wiring boards and certain other assemblies. Under this plan, a defective part is returned to General Electric and exchanged for a good part for a percentage of its purchase price.

Expedited delivery of parts is available in emergenecy situations. Contact your local General Electric sales office for more information.

REMOVING AND REPLACING PRINTED WIRING BOARDS

Use the following procedure if it becomes necessary to remove and replace a printed wiring board in an electronics module or field termination module:

1. Take the sequencer off line and turn off the DC power supply to the module.

2. Remove the board holder bracket on the front of the module and unplug any ribbon or I/O cables connected to the front of the board to be removed.

3. Remove the board by lifting out on the board extractors at the top and bottom of the board and sliding the board out. (Since the HRMB memory board contains a battery, if a memory board is removed from the sequencer module, data will be retained in the memory board for up to 10 days (at 25" C)).

4. If the board is to be replaced by a new board, remove the detachable daughter board (if any) from the old board by prying gently. Install the daughter board on the new mother board. The daughter board contains preset adjustable components and **must** be reinstalled on the new mother board whenever a mother board is replaced.

5. Replace the board in the proper slot, reinstall any ribbon or I/O cables, reinstall the board holder bracket and turn on the DC power supply to the module. Place the sequencer back on line and check for proper operation.

Printed wiring boards, if extremely dirty, can be cleaned in a lukewarm (100 degrees F) solution of mild household detergent or a laboratory detergent like ALCONOX or NEUTRA-CLEAN. Use a soft-bristled brush such as a paint brush to scrub the board. After cleaning, rinse thoroughly in several baths of lukewarm, fresh water. Shake off excess water, then air dry for a few hours.

MAINTENANCE AIDS

A number of maintenance and troubleshooting aids have been built into the DLC+ Sequencer and Programming Module. These include automatic system and board-level self tests, a Control Party Line test board, the ability to check sequencer operation via the programming module, and the ability to run system diagnostic tests. These maintenance aids are covered below.

HARDWARE SELF-TESTS

The DLC+ Sequencer and Programming Module hardware performs two types of automatic self-tests: a series of board-level self-tests and a system self-test when power is first applied. These self-tests are described in detail below.

Board-Level Self -Tests

All microprocessor-based boards in the sequencer and programming module (including the HMPC, HSHA, HMHA, and HCMA boards) perform automatic self-tests. These board-level self-tests continually check the operation of the board's microprocessor and associated circuits. As long as the board's microprocessor operates properly, a yellow LED labelled IMOK on the front edge of the board remains on. Should a problem develop, the IMOK LED goes out, allowing quick identification of a faulty board. The LIMA isolation module also contains a yellow IMOK LED, but this LED monitors the operation of the LIMA module's power supply.

System Self-Test

When either the sequencer or programming module is powered up, it goes through a system self-test. This self-test checks each board in the sequencer or programming module for proper operation. As the self-test executes, LEDs located on the programming module keyboard and sequencer HMPC board indicate whether or not a failure occurred. Should a problem be detected, the state of the LEDs can be used to help determine the faulty board. Since the programming module and sequencer self-test are different, they will be covered separately below.

Programming Module Self-Test. As shown in Figure 3-1, the three LEDs which monitor programming module self-test execution are located on the programming module keyboard. These LEDs are labelled READY, COIL, and ELEMENT. When the programming module is first powered up, all three LEDs will light for one second and then go out as the self-test begins. The LEDs then blink as described below while the self-test checks the major programming module boards in the following order:

HMPC board — The self-test first checks the CPU, RAM, PROM, and I/O circuits on the HMPC board. The COIL and ELEMENT LEDs blink once together for each circuit that passes the self-test (i.e., once for the CPU, once for the RAM, once for the PROM, and once for the I/O circuits). If these four circuits operate correctly, the READY LED comes back on and stays on during the remainder of the self-test

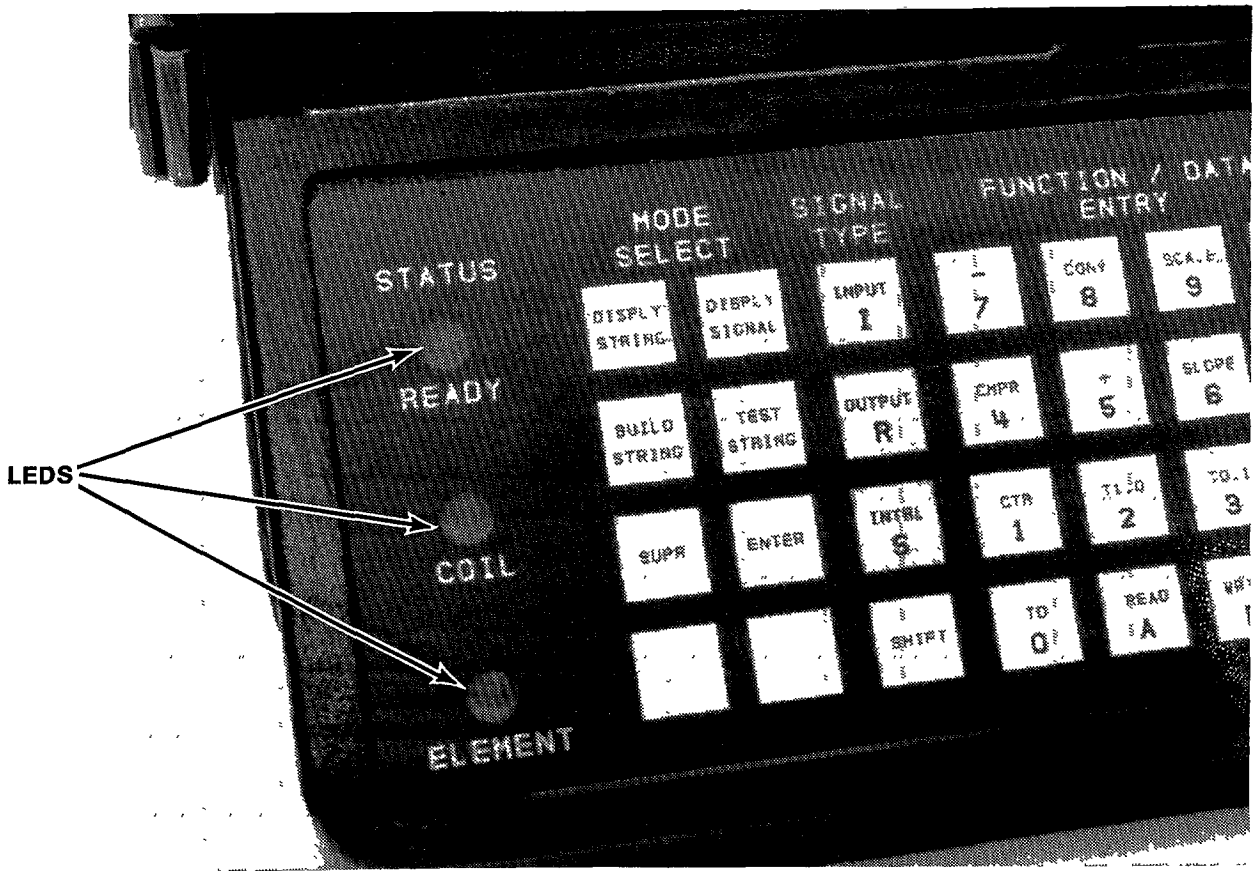
HPMA board — This subtest computes a checksum for the PROMs on the HPMA board and compares that checksum with the-original checksum written into the PROMs when they were manufactured. The subtest fails if the checksums do not agree. The COIL and ELEMENT LEDs blink once together during this subtest.

HRMB board — This subtest checks the operation of the RAM memory board in the programming module. Data is written to and read from each byte in memory in a non-destructive manner (i.e., data in memory is not destroyed). The COIL and ELEMENT LEDs blink once together during this subtest.

HVDA board — This subtest checks the operation of the video display circuitry in the programming module. The test takes about five seconds, during which time you will see a series of numbers and letters flash on the display screen. The COIL and ELEMENT LEDs blink once together during this subtest.

Note that after the initial one second blink at the beginning of the self-test, the COIL and ELEMENT LEDs come on at the start of a subtest and go out if the subtest passes. If an error occurs during the subtest, the COIL and ELEMENT LEDs will begin to flash rapidly (about ten times per second). By counting the number of blinks up until the time the rapid flashing begins, you can determine which board is malfunctioning. For instance, if the ELEMENT and COIL LEDs blink four times and then begin rapidly flashing, you will know that the HPMA board test has failed (i.e., four blinks indicate the HMPC board is OK; since the fifth blink began flashing, the HPMA test failed).

Once all of the subtests have executed succesfully, the programming module screen will display the message "PLEASE PRESS ANY KEY ON THE KEYBOARD". This message remains until you press a key, at which time the initial display will appear. After the system self-test has executed, the READY, COIL, and ELEMENT LEDs are used to monitor programming module status and the state of specific coils or contacts *(see* the ***Directo-Matic® Plus DLC+ Sequencer Operator Manual (GEK-73906)*** for more information).



(DSF-54326)

Figure 3-l.
Location of Self-Test LEDs on DLC + Programming Module

Sequencer Self-Test. The sequencer self-test is similar in operatidn to the programming module self-test except there is no LED that monitors subtest execution as the COIL and ELEMENT LEDs do. If you wish to monitor subtest execution, attach an oscilloscope to backplane pin PA67 (i.e., pin 67 on the backplane of the slot where the HMPC is located). The signal level on this pin during sequencer self-test execution acts exactly as the COIL and ELEMENT LEDs do during programming module self-test execution (i.e., the signal will toggle between high and low levels). The yellow IMOK LED on the front of the HMPC board acts exactly as the READY LED does during the programming module self-test.

When the sequencer is first powered up, the IMOK LED will light for one second and then go out as the self-test begins. The self-test checks the major sequencer boards in the following order:

HMPC board — This subtest first checks the CPU, RAM, PROM, and I/O circuits on the HMPC board. The signal level on pin PA67 toggles once for each circuit that passes the self-test (i.e., once for the CPU, once for the RAM, once for the PROM, and once for the I/O circuits). If these circuits operate correctly, the IMOK LED comes back on and stays on during the remainder of the self-test.

HPMA board — This subtest computes a checksum for the PROMs on the HPMA board and compares that checksum with the original checksum written into the PROMs when they were manufactured. The subtest fails if the checksums do not agree. The signal level on pin PA67 toggles once during this subtest.

HRMB board (s) — This subtest checks the opeiation of each RAM memory board present in the system (there must be at least one present). Data is written to and read from each byte in memory in a non-destructive manner (i.e., data in memory is not destroyed). The boards are tested in order of increasing address. In a PROM-based sequencer system, some HRMB boards may be replaced by HPMA boards. These extra HPMA boards are not tested by the sequencer self-test but may be tested by a diagnostic available through the programming module. The signal level on pin PA67 toggles once during this subtest.

HMHA board(s) — This subtest checks any HMHA boards strapped for hexadecimal addresses ODOOO to 0F000. The subtest fails if the HMHA board's own self-test has failed or if that portion of the board's RAM memory that is accessible to the sequencer system is faulty. The signal level on pin PA67 toggles once during this subtest.

If a board required in a minimum sequencer system is missing, its subtest will fail (a minimum system contains an HMPC board, an HPMA board, and an HRMB board strapped so that it includes hexadecimal addresses OCFOO to 0D000). If a board not required in a minimum system is missing (i.e., the HMHA board), its subtest will be bypassed.

If any subtest fails, the signal level on pin PA67 will toggle rapidly (about ten times per second) and the self-test will halt. By counting the number of times that PA67 toggles until the time the rapid toggling begins, you can determine which board is malfunctioning. For instance, if the signal on PA67 toggles slowly five times and then begins rapidly toggling, you will know that the HRMB board test has failed (i.e., four blinks indicate the HMPC board is OK and the fifth blink indicates that the HPMA board is OK; since the sixth blink began flashing, the HRMB test failed).

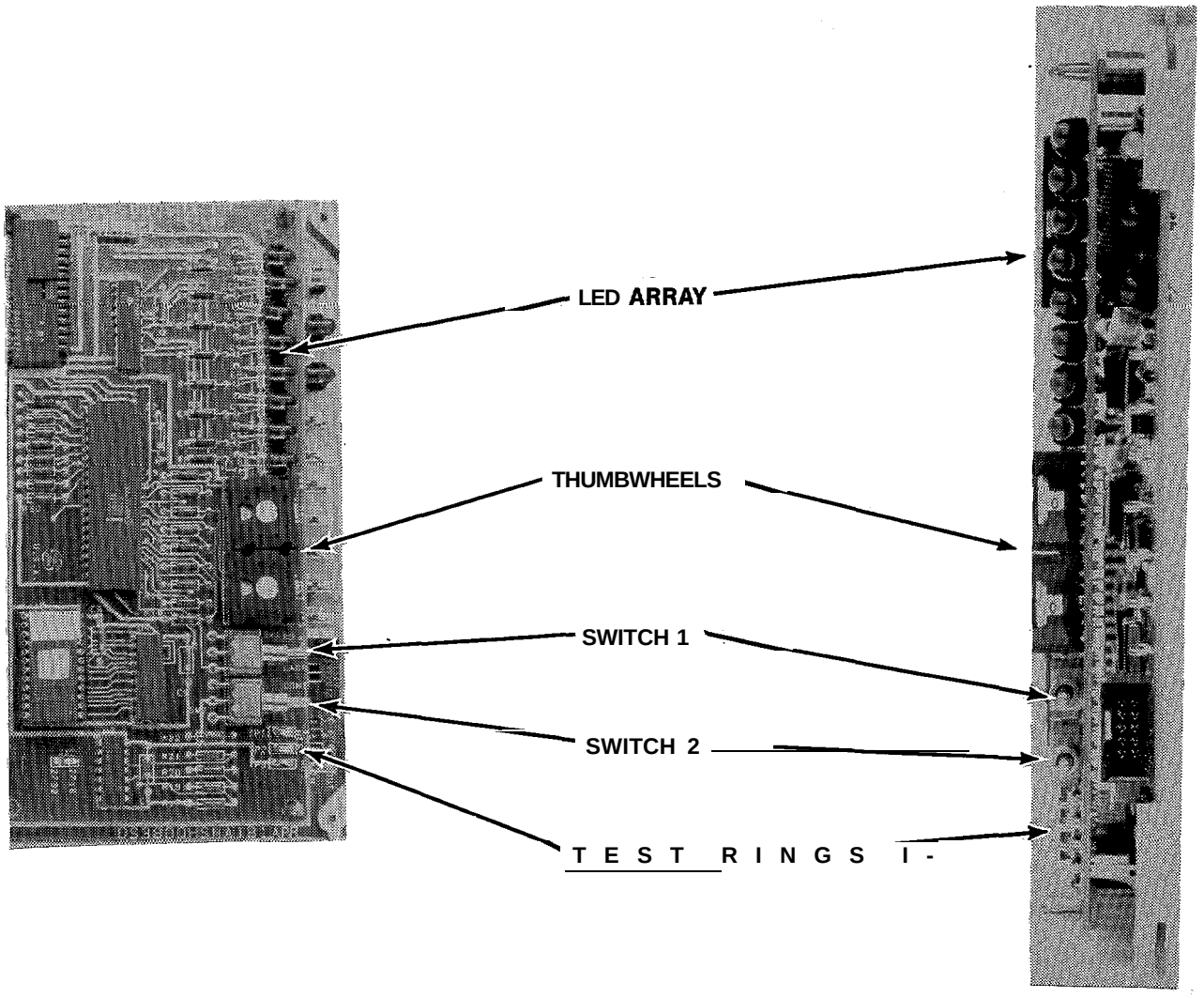
You may be able to diagnose some sequencer problems by connecting a programming module to the sequencer and placing the module into "Build String" mode. An error message indicating the problem may appear on the screen. Also, the sequencer can be tested using several diagnostics available through the programming module.

CONTROL PARTY LINE TEST BOARD (DSHA BOARD)

The DSHA daughter board allows an operator to test and monitor Control Party Line data flow through the HSHA slave highway board. Figure 3-2 shows the operator controls for the DSHA test board. Using the DSHA board, you can:

- Monitor inputs to and outputs from any I/O board connected to the HSHA
- Write data to any I/O board connected to the HSHA

- Logically disconnect the HSHA board and its associated I/O boards from the CPL link
- Monitor HSHA status (transmit and receive errors, CRC errors, etc.>



(DSF-54480 & 54472)

Figure 3-2.
CPL Test Board Switches and Indicators

The DSHA operator controls are described below.

LED Array - Displays I/O data, address, or status error counts depending on the position of Switch 1, Switch 2, and the configuration of the test rings. An address or status error count is displayed in binary form with the top LED as most significant bit and the bottom LED as least significant bit. I/O data is displayed in binary form with the bottom LED as most significant bit and the top LED as least significant bit.

Thumbwheels - Used to set up an address, enter I/O data, or select HSHA status errors to view. The top thumbwheel represents the most significant hexadecimal digit and the bottom thumbwheel represents the least significant hexadecimal digit. The setting of Switch 1 determines whether the numbers set in the thumbwheels represent an address, data, or status error selection. The address or data is not entered until Switch 2 is activated (this is true only in simulate mode).

Switch 1 - This three position switch controls the operation of the DSHA test board in conjunction with the test rings (described below). The lower position of the switch is momentary contact. Switch 1 causes different actions in each of the board's two test modes (monitor mode and simulate mode) as shown below:

Upper Position ("RD")

Monitor mode - Causes the LED array to continually display input data from the I/O board whose address has been entered into the thumbwheels. The input data is also available to the sequencer CPL link.

Simulate mode - Causes the LED array to display input data from the I/O board whose address has been entered into the thumbwheels. Switch 2 must be activated before this action occurs. The input data is **not** available to the sequencer CPL link.

Center Position ("WR")

Monitor mode - Causes the LED array to display output data sent to the I/O board whose address has been entered into the thumbwheels. The LED array will be updated with new data due to CPL activity.

Simulate mode - Causes the hexadecimal value set up in the thumbwheels to be written to an I/O board as an output. The I/O board address must first be selected by placing the test board into simulate mode, setting the desired address into the thumbwheels, placing Switch 1 into the lower "LADR" position (momentary contact), and then allowing Switch 1 to return to the center "WR" position. Placing Switch 2 into the lower "SS" position (momentary contact) or upper ("CONT") position writes the values set up in the thumbwheels to the desired I/O board.

Lower Position ("LADR") - momentary contact

Monitor *mode* - Causes the LED array to display HSHA status error information. The type of information displayed varies according to the setting of the lower thumbwheel (the upper thumbwheel must be set to 0) :

0 - Display number of HSHA transmit errors
1 - Display number of HSHA receive errors
2 - Display number of CRC (Cyclic Redundancy Check) errors
3 - Display number of overrun errors
4 - Display number of abort errors
5 - Display number of passes (least significant byte)
6 - Display number of passes (middle byte)
7 - Display number of passes (most significant byte)
8 - Display number of slave errors

Simulate mode - Causes the values set in the thumbwheels to be used as an I/O board address. This must be done prior to writing outputs to an I/O board in simulate mode.

Switch 2 - This three position switch activates the function set up by Switch 1. The lower position of the switch is momentary contact. Switch 2 is active only in simulate mode and causes the following actions:

Upper Position ("CONT")

Causes the function set up by Switch 1 to be performed continually.

Center Position

Turns off the function set up by Switch 1.

Lower Position ("SS") - momentary contact

Causes the function set up by Switch 1 to be performed only once (single shot function).

Test Rings - Determines the test board's mode of operation. The board normally operates in monitor mode, meaning that the HSHA and I/O boards respond normally to the CPL link. On-line monitoring of I/O data and status errors can occur in this mode. Jumpering SIM1 to COM causes the test board to operate in simulate mode, which causes the HSHA and all of its I/O boards to be logically (not electrically) disconnected from the CPL link (that is, the HSHA and I/O boards respond only to the test board, not to the CPL software). In simulate mode, data can be sent to particular I/O boards under control of the test board. Test ring SIM2 is not currently used.

Instructions for using the DSHA board to test and monitor CPL operations are included below.

1. TO MONITOR DATA INPUT FROM AN I/O BOARD TO THE SEQUENCER:
   a. Remove jumper (if any) between SIM1 and COM test rings.
   b. Position Switch 1 into the "RD" (upper) position.
   c. Select the desired I/O board address using the thumbwheels.
   d. Data being input from the board will appear in the LED array (most significant bit at the bottom, least significant bit at the top).

2. TO MONITOR DATA OUTPUT TO AN I/O BOARD FROM THE SEQUENCER:
   a. Remove jumper (if any) between SIM1 and COM test rings.
   b. Position Switch 1 into the "WR" (center) position.
   c. Select the desired I/O board address using the thumbwheels.
   d. Data being output to the board will appear in the LED array (most significant bit at the bottom, least significant bit at the top).

3. TO DISPLAY HSHA STATUS ERROR INFORMATION:
   a. Remove jumper (if any) between SIM1 and COM test rings.
   b. Position Switch 1 into the "LADR" (lower) position (momentary contact).
   c. Use the thumbwheels to select the type of status error information to be displayed (see description of Switch 1 above for more information).
   d. Status error information will appear in the LED array (most significant bit at the top, least significant bit at the bottom)
   e. To return to the normal monitoring mode, place Switch 2 into the lower ("SS") position (momentary contact).

4. TO LOGICALLY DISCONNECT THE HSHA BOARD FROM THE CPL LINK AND DISPLAY DATA INPUT FROM AN I/O BOARD:
   a. Jumper test ring SIM1 to COM.
   b. Position Switch' 1 into the "RD" (upper) position.
   c. Select the desired I/O board address using the thumbwheels.
   d. Position Switch 2 to select continuous read, single shot read, or no action. Data being input from the board will appear in the LED array (most significant bit at the bottom, least significant bit at the top).

5. TO LOGICALLY DISCONNECT THE HSHA BOARD FROM THE CPL LINK AND WRITE OUTPUT DATA TO AN I/O BOARD:
   a. Jumper test ring SIM1 to COM.
   b. Select the desired I/O board address using the thumbwheels and press Switch 1 down into the "LADR" position to load the address. The address will appear in the LED array (most significant bit at the top, least significant bit at the bottom),
   c. Position Switch 1 into the "WR" (center) position and select the desired output byte pattern with the thumbwheels.
   d. Position Switch 2 to select continuous write, single shot write, or no action.

PROGRAMMING MODULE

The DLC+ Programming Module has several features which aid in troubleshooting and maintaining the DLC + Sequencer, including:

- The ability to force signals. This feature allows you to force an output signal into an energized or deenergized condition and then check to make sure the applicable output device responds accordingly. Since the signals are forced at the application software level, this feature checks the proper operation of the CPU, memory, and communications hardware as well as the I/O board itself.

- The ability to check input and output signal status. This feature allows you check whether an I/O signal is energized or deenergized at the application program level.

- The ability to detect sequencer error conditions and report them as error messages. Should the sequencer malfunction while the programming module is connected, an error code and explanatory message appear at the bottom of the programming module screen when the module is in "Build String", mode.

- The ability to run diagnostics. Using the programming module, you can run sequencer CPU, memory, and communication diagnostic tests. These tests are described in the next section of this manual.

All of these features are explained in greater detail in the ***Directo-Matic® Plus DLC+ Sequencer Operator Manual (GEK-73906)***.

DIAGNOSTIC TESTS

A number of diagnostic tests are available via the Supervisory mode of the DLC+ Programming Module. These tests exercise the memory, processor, and communication channels of the DLC+ Sequencer. The operating procedures for these tests are found in the ***Directo-Matic® Plus DLC+ Sequencer Operator Manual (GEK-73906)***. You must take the sequencer off-line before running any of these tests. The tests and the functions they exercise are summarized below.

Memory Test

This diagnostic tests a specified memory range to determine whether the RAM memory (HRMB boards) in that range can be written to and read from. Application software in memory is not damaged by this routine. The test produces a display showing the status (GOOD or BAD) of adjacent 4 kilobyte sections of memory.

Bus Hog Test

This test determines the type of memory present in the sequencer (RAM, ROM, or none) and the address range of each type. A bus hog is reflected as a "no memory" condition. Application software in memory is not damaged by this routine. The test produces a display showing the address range of each type of memory.

CPL Response Test

This test determines which stations on a CPL channel respond to a trasmission from the HMPC board. After you enter the desired CPL channel number, the test produces a display showing the station numbers that have responded. The test is used to determine the I/O configuration (number of stations) of a particular channel.

CPL Echo Test

This test determines which stations (configured in the KIB) on a CPL channel are echoing transmitted data correctly. After you enter the desired CPL channel number the test begins running. The test produces a display showing each station number, the total number of responses from that station, the number of responses in error, and other information that would help identify possible malfunctions. This test loops continuously until stopped by the user.

CPL Status Test

This diagnostic tests a specific station on a specific channel and displays a variety of possible status errors: transmit, CRC, station, and receive errors. This test loops continuously until stopped by the user.

CPL Channels Reset

This routine allows the user to reset a specific channel on an HMHA. There are two channels on each HMHA.

HMPC Test

This is the same test that is automatically performed when the sequencer is first powered up. The test checks the CPU, ROM, RAM, and I/O on the HMPC and displays an error message if an error occurs.

OTHER MAINTENANCE AIDS

There are several other sequencer maintenance aids not described above. These include:

- **Test points** on most printed circuit boards; these test points are described in the backsheets for each board.
- A **bus isolation switch** on each I/O board which effectively isolates all bus inputs and outputs on that board from P5 (+5 VDC) or DCOM (Common). If a system bus appears to be shorted to P5 or DCOM, the bus isolation switch on each board can be toggled to determine which board contains the short. The switch should be up for normal (on-line) operation and down for bus isolation.
- **Signal status LEDs** (typically eight or sixteen red LEDs) on the front edge of all I/O boards indicate the state of each input or output signal on the board; a lighted LED indicates an active I/O signal. In addition, another LED (yellow) indicates whether the data bus parity is correct; when lighted, this LED indicates correct read parity (for input signals) or write parity (for output signals).
- **Blown fuse indicators** are located on most boards that contain fuses; blown fuses can be identified and replaced without removing the board from its card rack.

NOTES

CHAPTER 4
TROUBLESHOOTING

This chapter contains troubleshooting procedures for the DLC+ Sequencer and DLC+ Programming Module. Only the most common problems are covered in this chapter. A good understanding of the theory of operation of the programming module and sequencer will help you troubleshoot problems in those devices.

When troubleshooting problems, it is best to look at the programming module and sequencer as a system composed of six distinct parts: the programming module, the processor, the power supply, the communications interface, the I/O boards, and the I/O devices ("real world" devices such as solenoids, operator controls, etc.). This concept is shown in Figure 4-l. When the system is divided up this way, it is much easier to isolate problems.

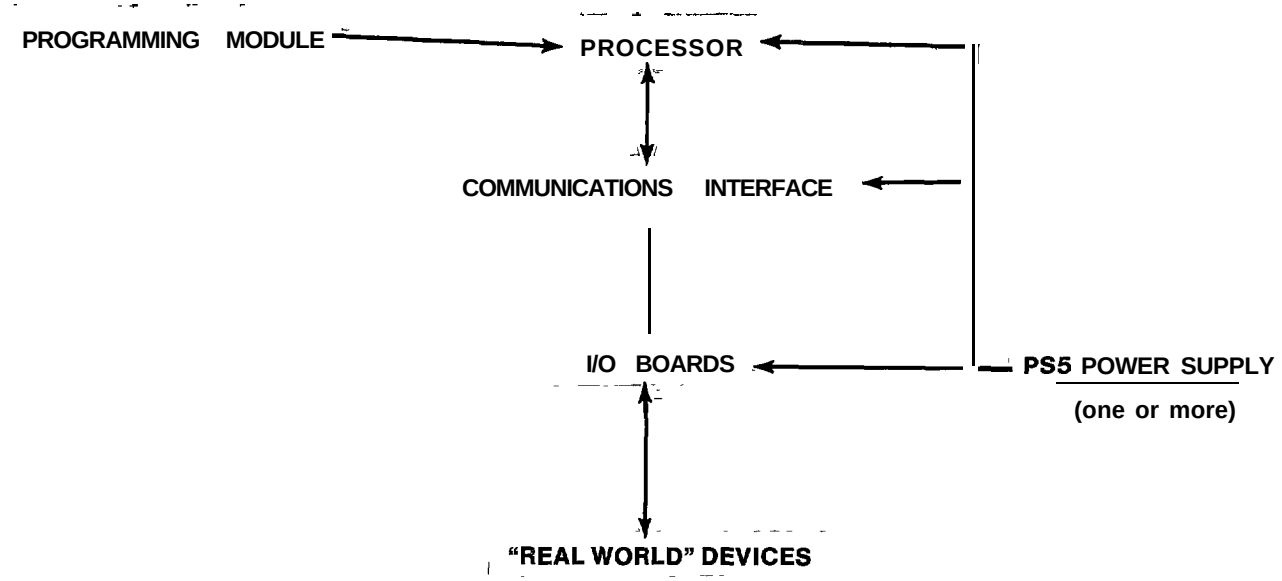


Figure 4-l.
"System" Concept of Troubleshooting

You can troubleshoot most **programming module** problems using the built-in self-test as described in the next section. Since the programming module is used to troubleshoot the sequencer, make sure the programming module works properly before doing any other troubleshooting.

The **processor** also contains a built-in self-test that aids in troubleshooting. Some troubleshooting can be done by simply observing the IMOK light on the front edge of the HMPC board, but isolation of faulty boards requires that a programming module be connected to the sequencer so that error messages can be displayed.

The **power supply** consists of one or more PS5 DC power supplies, each of which contains an IMOK light that indicates whether the outputs from the supply are within proper voltage range. Power supply problems are usually fairly obvious, but it may be necessary to measure the outputs of each supply to determine which PS5 unit is at fault.

The **communication interface** may consist of backplane wiring (for module I/O>, ribbon cable (for local I/O>, or a series of HMHA and HSHA boards connected by twisted pair wire (for remote I/O via CPL). Troubleshooting the CPL hardware is simplified by board level self-tests and diagnostics.

Most **I/O boards** have signal status lights on the front edge that allow monitoring of incoming or outgoing signals. Using these lights in conjunction with the forcing capability of the programming module, you can easily determine whether an I/O fault is caused by one of the I/O boards or one of the I/O devices.

I/O devices such as solenoids, relays, limit switches, operator controls, and motor starters are more likely to fail than sequencer components.

When a fault occurs, try to determine which of the above six "blocks" would most logically be at fault. For instance, if one I/O device is not working properly, you can assume that the device or its I/O board is at fault. If a whole rack of remote I/O devices is not working, the problem could be within the CPL hardware or the PS5 power supply for that rack. If the sequencer is not working at all, the problem could lie in the processor or the PS5 power supply for the processor. Once the correct "block" is determined, troubleshoot that "block" using this troubleshooting guide and the maintenance aids described in Chapter 3.

TROUBLESHOOTING THE PROGRAMMING MODULE

Plug the programming module power cord into a known good 115VAC outlet and attach the proper signal cable between the programming module HMPC board and the sequencer HMPC board. Turn on the module using the power switch on the back. After a short self-test, during which the status lights should flash and the screen should display a shifting array of characters, the message "PLEASE PRESS ONE OF THE KEYS ON THE KEYBOARD" should appear on the screen. Refer to the information below if the programming module does not work correctly.

Problem: **Programming module screen does not light up and fan is not running.**

Cause: No AC input to module.

Action:
1) Verify that the AC input voltage is within proper limits (115VAC ± 10 volts) and that the AC receptacle has been properly wired (hot line and ground).
2) Check that the line fuse located near the power switch is good.
3) If the above two steps check out satisfactorily, there could be an internal AC power problem in the module or the fan could be defective. The module should be serviced by General Electric Company.

Problem: **Image on screen *is* not stable.**

Cause: Video horizontal/vertical hold control is not working properly.

Action: Shut power off to the programming module and unplug the HVDA board in the back of the module. Wait 30 seconds and then reinsert the board and reapply power to the module. Repeat this procedure several times if required. If this does not help, the module should be serviced by General Electric Company.

Problem: **Just after power-up, the yellow READY status light goes off** (or **never comes on). The red COIL and ELEMENT status lights may begin to blink rapidly (10 times per second).**

Cause: The automatic self-test (described in Chapter 3) has detected an error in one of the module's printed wiring boards.

Action: Turn the module off and then back on. Watch the status lights and follow the directions below:

If the yellow READY light does not come on at all, replace the HMPC board.

If the READY light comes on and the two red lights immediately begin flashing, replace the HPMA board.

If the READY light comes on and the two red lights blink once and then begin flashing, replace the HRMB board.

If the READY light comes on and the two red lights blink twice and then begin flashing, replace the HVDA board.

Problem: **Sporadic error messages displayed while the programming module is in the build string or display string mode.**

Cause: The programming module may need to be reset, or the sequencer sweep rate may be too fast, providing insufficient time for the programming module control software to work.

Action: **1)** Turn the programming module off and then back on.

2) If this did not work, check the sweep rate via the KIB "System Parameters" function in the Supervisory menu to ensure the sweep rate has not been changed. The programming module must be connected to a properly operating sequencer before checking the sweep rate.

Problem: **Error code appears on the programming module screen while the module is connected to an operating sequencer.**

Cause: Fault in the sequencer or programming module.

Action: Refer to the list of error codes in Appendix B to determine what board probably caused the error message. If the error code appears in the bottom left corner of the screen, the problem is within the sequencer. If the error code appears in the bottom center of the screen, the problem is within the programming module.

TROUBLESHOOTING THE SEQUENCER

Power up and check out the sequencer and I/O modules as described in Chapter 3 in the section titled "Initial Set-up and Check-out of Equipment". Refer to the information below if the sequencer does not work correctly.

Problem: **IMOK light on PS5 power supply does not come on when the power supply circuit breaker is turned on. DC power is not being supplied to the sequencer and/or I/O modules.**

Cause: Power supply output voltages not within acceptable range.

Action: **1)** Turn the circuit breaker off for about 5 seconds and then back on to reset the power supply. Try this several times.

2) If the IMOK light still remains off, measure the output bus voltages (P28V, P5V, P15V, N15V) **at the load** and adjust the voltages if necessary Adjust P28V to +28VDC ±VDC with pot V2. Adjust P15V to +15VDC ±.15VDC with pot V3. Adjust N15V to -15VDC ±.15VDC with pot V4. If the P5V bus is less than +4.8VDC, remove the loads from the power supply and reset the power supply. If the IMOK light now comes on and the P5V voltage is +5 .O ±.1VDC, the P5V voltage source is being overloaded. If the voltage remains low, reconnect the loads and adjust the P5V voltage to +5.0 ±.1VDC with pot V1. If the voltage will not adjust properly, replace the power supply.

Problem: **Sequencer intermittently reacts as though DC power has been** lost.

Cause: Intermittent switching of the LVSL power failure signal from the NPSF board mounted on the PS5 power supply.

Action: This problem can be caused by excessive AC waveform distortion. Verify that the berg jumper on the NPSF board is in the "ACDIS" position. Also ensure that incoming AC power to the PS5 supply is within specifications. If both of the above check out, replace the NPSF board and/or the PS5 power supply.

Problem : **IMOK light on the HMPC board does not come on when the sequencer is first powered up or goes off during operation.**

Cause: The HMPC board has failed the power-up self-test or has failed during operation.

Action: Follow these steps until the problem is corrected:

1) Attempt to manually reset the HMPC by momentarily jumpering test ring TP1 to TP2 (DCOM) on the board's front edge.

2) Verify that the berg jumpers on the HMPC board are properly configured according to the system elementary.

3) Turn off power to the sequencer, then turn power back on. If the IMOK light does not come on, replace the HMPC board.

Problem : **Sequencer will not operate when first powered up. HMPC IMOK light is on.**

Cause: The sequencer self-test has probably detected an error.

Action: **1)** Turn power off to the sequencer and connect an oscilloscope probe to backplane pin PA67 (pin 67 on the backplane part of the sequencer HMPC card slot). Turn power back on and determine which board is failing by noting when the signal on PA67 begins to rapidly toggle This procedure is described under "Sequencer Self-Test" in Chapter 3.

2) Connect a working programming module to the sequencer (via the jack on the HMPC board) and place the module in Build String mode. Observe any error codes that appear at the bottom left of the screen. Appendix B lists these codes and tells which boards may be faulty for each code. Since error code 21 or 22 could indicate bad data in an **HRMB** board, try clearing and reloading memory using the programming module; if this does not solve the problem, one or more of the HRMB boards will have to be replaced.

3) Use the programming module to run diagnostic tests on the sequencer. These tests are described in Chapter 3 of this manual and in the ***Directo-Matic® Plus DLC+ Sequencer Operator Manual (GEK-73906)***.

Problem: **Yellow IMOK light on HMHA board is off.**

Cause: The board's self-test has detected an error.

Action: 1) Turn power off to the module containing the HMHA and then back on to reset the HMHA.
2) Verify that all berg jumpers are correctly configured according to the system elementary.
3) Verify that the PS5 power supply for the module containing the HMHA board is operating properly and that the power cable between the power supply and module is tightly connected.
4) Measure the differential voltage across the IMOK LED to determine if the LED has failed.
5) If the IMOK light remains off, replace the board.

Problem: **Red lights on HMHA board stay off after power is applied, but the yellow IMOK light is on.**

Cause: The HMHA board communicates with one or two remote HSHA boards over twisted pair wires to implement the Control Party Line data link. Each of the two red lights on the HMHA board monitors communication traffic between the HMHA and a remote station. Assuming that both channels are being used, if either of the red lights begins to blink, there has been a communication failure with a CPL remote station.

Action: 1) Ensure that the PS5 power supply for the remote station(s) is on and that the signal cables into the HSHA and HMHA boards are making good contact.
2) If LIMA isolators are not used, check the continuity of the twisted pair cable between the HSHA and HMHA board of the affected channel. If one or more LIMA isolators are used, check the LIMA modules as described in the next section of this troubleshooting procedure. Also check all signal connections between the LIMA modules and the HSHA and HMHA boards.
2) Ensure that the berg jumpers on the HMHA board are configured according to the system elementary.
3) Use the programming module to check the sweep rate via the KIB "System Parameters" function in the Supervisory menu to ensure the sweep rate has not been changed. A too-slow sweep rate can cause apparent loss of communications.
4) Use the programming module to run applicable CPL diagnostics. The results may indicate the need to replace the HMHA or HSHA boards.

Problem : **Yellow IMOK light on LIMA module is off.**

Cause: Failure of isolated P5V voltage to module.

Action: 1) Check that the power input to connector JP is correct per the elementary diagram. Measure the voltages at the back of the module. If the voltages are not correct, trace the problem back to the module where the voltages originated.
2) If these voltages appear correct, measure the isolated P5V voltage between PA pins 4 (P5V) and 2 (COM). If the voltage is not +5VDC ±.1VDC, replace the HIMA board in the module. If the voltage reads correctly, the IMOK LED is probably defective.

Problem : **Bad data sent between HMHA and HSHA boards via LIMA modules; all IMOK and status LEDs are OK.**

Cause: Bad connection or defective LIMA module.

Action: 1) Check that the signal connections into the LIMA modules are tight.
2) Verify that the LIMA modules at both ends of the communications link have the J1 and J2 berg jumpers in the "IN" position. If there are any intermediate LIMA modules on the communications link, their J1 and J2 berg jumpers should be in the "OUT" position.
3) Replace the HIMA board in each LIMA module one at a time and see if the problem has been solved. If not, replace each LIMA module one at a time until the problem has been solved.

Problem: **Yellow IMOK light on HSHA board is off.**

Cause: The board's self-test has detected an error.

Action: 1) Turn power off to the module containing the HSHA and then back on to reset the HSHA.
2) Verify that all berg jumpers are correctly configured according to the system elementary.
3) Verify that the PS5 power supply for the module containing the HSHA board is operating properly and that the power cable between the power supply and module is tightly connected.
4) Measure the differential voltage across the IMOK LED to determine if the LED has failed.
5) If the IMOK light remains off, replace the board.

Problem: **Red LACT light (second from top) and/or ADR light (third from top) on HSHA board does not come on during attempted communication between the HMHA and HSHA boards. All applicable IMOK lights are on.**

Cause: When on, the LACT light indicates that the Control Party Line is active and the ADR lights indicates that the HSHA is being addressed by the CPL. If either of these lights does not come on during attempted communication, the communication line (twisted pair and LIMA modules), HSHA board, or HMHA board could be bad.

Action: 1) Check all connections along the communications link between the HMHA and HSHA boards. Check the continuity of the twisted pair cable between connection points.
2) Verify that all HSHA berg jumpers are properly configured according to the system elementary
3) Measure the differential voltage across the LACT LED to determine if the LED is faulty
4) If none of the above steps work, replace the following boards or modules in this order:

1. HSHA
2. HIMA (All applicable)
3. HMHA
4. LIMA (All applicable)

Problem: **Yellow parity light on one or more I/O boards is off.**

Cause: Input or output parity error or defective I/O board.

Action: 1) Ensure that the bus isolation switch on the suspect board is in the normal (up) position.
2) If the yellow parity light on any other boards is on, measure the differential voltage on either side of the yellow parity LED on the suspect board. If there is + 5VDC across the LED and the LED does not come on, the LED is defective.
3) Swap the suspect board with another like board whose yellow parity light is on. This will determine whether the board or the card slot is bad.

Problem: **Red input/output monitoring LEDs on I/O boards do not respond to input or output signals.**

Cause: When an input or output signal through a sequencer I/O board becomes active, the associated LED on that I/O board should illuminate. Failure of this to occur indicates an I/O device or board problem.

Action: 1) Ensure that the bus isolation switch on the suspect board is in the normal (up) position.
2) Ensure that the I/O signal cables into the board or module are tightly connected.
3) Measure the differential voltage on either side of the affected LED(s) on the suspect board. If there is +5VDC across the LED and the LED does not come on, the LED is defective.
4) If the suspect board is an output board, use the programming module to force the related relay string active and note the results. If the suspect board is an input board, use the programming module to monitor the input relay contact while forcing the related external device active. Using this method, it should be possible to tell whether the fault lies in the sequencer program, the I/O board, or an external device.
5) Swap the suspect board with another like board whose red LEDs work properly. This will determine whether the board or the card slot is bad.

Problem: **Output device does not operate, even though red monitoring LED on** I/O **board does light.**

Cause: Bad I/O connection, bad device, or blown fuse (on some types of I/O boards).

Action: Check for a blown fuse or bad I/O connection on the I/O board. Ensure that the device (i.e., solenoid, switch, etc.) is not faulty.

APPENDIX A
PRINTED WIRING BOARD DESCRIPTIONS

This section of the maintenance manual contains brief descriptions of the most common DLC+ Sequencer printed wiring boards. These boards are of two types: sequencer boards and I/O boards. Sequencer boards include microprocessor, memory, communications, and other boards involved with the logical operation of the sequencer. I/O boards include boards used for signal conditioning of input and output signals. The boards are listed in alphabetical order under each type. Daughter boards are listed directly after their associated mother board. Some boards are considered "modules", which means that they are general purpose boards containing PROM chips that give the board its particular function. Only the highlights of each board are presented; for more detail, consult the Directo-Matic @Plus backsheets for the particular board.

Only those Directo-Matic ® Plus boards currently associated with the DLC + Sequencer module are listed here. New boards may have been added since this manual was written. Check with your General Electric Company sales representative for up- to-date information.

SEQUENCER BOARDS

HACA (Address Customizing Board) Consists of wire wrap pins for customized addressing for up to 18 I/O printed wiring boards in an electronics module (TEM, SEM, or MEM) . Addresses within the module can be specified by wire-wrapping appropriate pins together. Berg jumpers can be configured to define sense, freeze, reset, and I/O or slave mode logic for the module.

HACB (Address Customizing Board) A 5 3/4" board which is housed in a card box in the horizontal I/O stick (DS3820HIO). The board contains wire wrap pins for customized addressing for up to 8 I/O printed wiring boards in one field termination module (horizontal stick). Addresses within the module can be specified by wire-wrapping appropriate pins together. Berg jumpers can be configured to define sense, freeze, reset, and I/O or slave mode logic for the module.

DACB (HACB Daughter Board) Enables HACB board to provide customized addressing to 8 I/O printed wiring boards in a VTDD.

HBDA (Buffer Decoder) Interfaces address, data, and control signals between the slave controller and I/O boards in a CPL slave station. Seven logic inputs and outputs can be accessed from the data bus. The HBDA is strapped into one of eight address ranges and can handle up to 16 work addresses in the range.

HCMA (Dual Communications Interface) An 8085 microprocessor-based dual channel serial communication subsystem. It provides an interface to two RS232 full or half duplex serial communication channels. The channels can be programmed for synchronous or asynchronous baud rates up to 19.2 K bits/sec. It accepts either a 2K or 4K byte EPROM to program communication protocol.

HIMA (CPL Isolation Board) Provides optical isolation for CPL master and slave stations. The HIMA board will drive up to 4000 feet of twisted pair wire and is used in the LIMA module.

HIOB (Siltron™ CPL Slave I/O) Interfaces the SILTRON™ drive to the Control Party Line as a CPL slave station.

HKIB (Keyboard Input) A single-chip microcomputer board programmed to monitor a switch-matrix (57 contacts) at 10 ms. intervals. It encodes into 8 bits parallel and generates a 5 microsecond strobe. It interfaces and decodes the keyboard of the DLC+ Programming Module.

HLSC (Link Slave Control) A slave device used for information exchange on a serial data link.

HLSD (Link Slave Control) A revised edition of the HLSB that replaces it totally in function. It is used for information exchange on an eight bit parallel data bus, including interrupt generation and handling.

HLXA, HLXB (CSH Modulation-Demodulation) As part of a CSH slave station, these cards provide modulation and demodulation for interface to the Control Signal Highway. They are used with various other card clusters to implement CSH master, slave and test functions.

HMHA (Master Highway Controller) An 8085 microprocessor-based dual channel serial communication subsystem. It has two SDLC/RS422 party line serial channels that can be programmed for baud rates up to 48K bits/sec, synchronous. It interfaces to the L-Bus via a 2K byte dual port RAM. It accepts either a 2K or 4K byte EPROM to program communication protocol.

HMPC (Microprocessor) The heart of the DLC+ Sequencer, consisting of sequencer product software (in PROM) and a complete microcomputer including 4K bytes of RAM memory and provision for 16K bytes of PROM memory, plus an RS-232 communications interface. The board communicates with other boards (memory, I/O, and communication) via the L-bus. The HMPC board, with different product software, is also used in the DLC+ Programming Module.

HPBA, HPBC (Parallel Line Buffer) Interfaces address, data, and control signals between the CPL slave module and I/O boards. Has two modes of operation: (1) Buffer-Drive mode passes one-for-one all of the F-level signals to the R-level, making a new F-level driving source; (2) Buffer decode mode provides full decoding for up to 32 word addresses. HPBA also checks all of the R-level signals for shorts to P5 or DCOM.

HPMA (32K/64K PROM Board) This board contains up to 64K bytes of Programmable Read Only Memory (PROM), which may contain product or application software.

HRMB (8K Bytes Battery-Backed RAM (BRAM) Memory) Provides 8K bytes of battery-backed CMOS RAM memory for storage of application software. The HRMB board is dual-ported to both the E-bus and L-bus. In case of power failure or removal of the board from an electronics module, the board will retain memory data for up to 8 days (depending on battery charge and temperature). The on-board NiCad battery is automatically charged whenever the DLC+ Sequencer is operating.

HSHA (CPL Slave Highway) Operates as a CPL slave station when connected to a CPL master station via twisted pair wires. Interfaces I/O boards to the Control Party Line.

DSHA (Optional HSHA Daughter Board) Provides LEDs, thumbwheels, and toggle switches for monitoring data flow through the HSHA mother board and performing off-line testing of any I/O board connected to the HSHA.

HVDA (Video Display Interface) Provides an interface between a CRT monitor and the HMPC board and L-bus. This board provides both TTL and composite video signals.

HXPB (HMPC Expander Board) An expander board for the HMPC microprocessor board which provides the following L-bus functions: 4K RAM memory, 4 PROM sockets, digital I/O via front edge connector, 8 interrupts, counters, and a programmable stall timer.

LIMA (CPL Isolation Module) Consists of an HIMA board and a standard telephone jack in a special electronics module for panel mounting and termination of signal cables.

PCLA (CPL Master Station Module) This module consists of an HMHA board plus a PROM chip that enables the board to act as a CPL master station. The PCLA interfaces remote I/O devices via HSHA or HIOB boards to the DLC+ Sequencer over dual twisted pair transmission lines. Communication can be point-to-point or multipoint, and either non-isolated (for distances less than 30 feet) or isolated via the LIMA module (for distances up to 4000 feet>.

PRBA (CPL Remote Buffer Module) This module provides a buffer and interface between a host microprocessor and the CPL master station (PCLA). This enables the host microprocessor to communicate with other processors (which look like CPL slaves) over the Control Party Line.

I/O BOARDS

HLIA (Logic (+5VDC) Input) Provides an interface between 16 logic inputs and an S-bit data bus. The 16 inputs are filtered for noise immunity. Sixteen LEDs display the status of the inputs. Bus access to the high and low bytes and the high impedance state are controlled by other digital inputs. Accepts 16 logic level (+ 5VDC) signals as non-isolated inputs to the DLC + Sequencer.

HLIC (Logic (+ 5VDC) Input) Accepts 8 logic level (+ 5VDC) signals as isolated inputs to the DLC+ Sequencer.

HLOA (Logic (+ 5VDC) Ouput) Provides buffering, parity check, and LED indication of 16 logic non-isolated inputs from the DLC+ Sequencer. The outputs may be enabled (from the high impedance state) by address decoding logic. The outputs are available both on the backplane and through a front edge connector.

HLOB (Logic (+5VDC) Output) Provides 8 logic level (+5VDC) signals as non-isolated outputs from the DLC + Sequencer.

HRRB (Reed Relay Output) The card is used to provide 8 normally open reed relay contacts as isolated outputs from the DLC + Sequencer.

HSCA (Signal Conditioning (115VAC, +105VDC) A 115 VAC or 105 VDC isolated signal conditioning card consisting of eight high level optically isolated input circuits, logic output drivers, drive level interface, and a bus isolation circuit. The eight inputs are available after conditioning as logic output signals for direct use or can be read via the drive level interface. Accepts 8 high level signals (115VAC or + 105VDC) as isolated inputs to the DLC + Sequencer.

HSCB (Signal Conditioning (+125VDC)) Accepts 8 high level signals (+125VDC) as isolated inputs to the DLC + Sequencer.

HSCC (Signal Conditioning (+28VDC)) Accepts 16 high level signals (+28VDC) as non-isolated inputs to the DLC+ Sequencer.

HSCD (Signal Conditioning (+105VDC)) A 105 VDC non-isolated signal conditioning card consisting of sixteen filtered 105V inputs, logic output drivers, drive level interface, and a bus isolation circuit. The input signals are available after conditioning as logic output signals for direct use or can be read via the drive level interface. Used as non-isolated inputs to the DLC + Sequencer.

HSCE (Signal Conditioning (+ 105VDC)) Accepts 8 high level signals (+ 105VDC) as non-isolated inputs to the DLC + Sequencer.

HSCF (Signal Conditioning (+ 28VDC)) Accepts 8 high level signals (+ 28VDC) as non-isolated inputs to the DLC + Sequencer.

HSDA (Solenoid Driver) Provides 4 fused 115/230VAC solenoid driver signals as isolated outputs from the DLC + Sequencer.

HSDC (Solenoid Driver) Provides 4 non-fused 115/230VAC solenoid driver signals as isolated outputs from the DLC+ Sequencer.

HSDD (Solenoid Driver) Provides 8 fused 115/230VAC solenoid driver signals as isolated outputs from the DLC + Sequencer. Includes blown fuse indicators. Fuses may be changed without removing the card from the electronics module.

NADA (Analog to Digital Converter) Provides 12 bit analog to digital conversion for one of eight differential input channels (+ 12.5VDC full scale) selected by the HMPC board.

NDAC (Digital to Analog Converter) A 12-bit binary bipolar analog-to-digital converter (11 bits data plus sign) on a single size board. Includes an 8 channel static analog scanner to accept up to eight ±10 volt analog signals. Requires externally supplied clock (accepts 2.045 MHz normally from HMPA) . Converted data can be ready 10 microseconds after providing address of point to be converted. Converted output is available externally on buffered data bus.

APPENDIX B
SELF-TEST ERROR CODES

Both the DLC+ Sequencer and the DLC+ Programming Module perform automatic self-tests during initialization. Should a DLC+ Sequencer self-test fail, the IMOK light on the affected board will go out. An error message is displayed on the programming module screen if the module is attached to the sequencer and is in Build String mode. Should a DLC+ Programming Module self-test fail, the IMOK light on the affected board will go out, but an error message is not displayed unless the module is performing its power-on self-test. In either case the error message reads SELF TEST ERROR NN, where NN in one of the error codes listed below.

If the error is in the programming module, the error code appears at the bottom center of the screen. If the error is in the sequencer, the error code appears at the bottom left of the screen along with an error explanation.

ERROR CODE	MEANING
00	HMPC RAM failure
01	HMPC product software checksum error
03	HMPC USART failure
04	HMPC interrupt controller failure
05	HMPC timer failure
06	HMPC bus hog failure
11	HPMA product software checksum error
21	HRMB/HPMA application software checksum error
22	HRMB/HPMA application software initialization failure strings cleared from memory
30	HMHA RAM failure (Remote I/O)
31	HMHA self-test failure (Remote I/O)
32	HMHA configuration error (Remote I/O)
40	HVDA RAM failure
50	Application software database consistency check
90	No memory available to complete initialization
91	Out of I/O memory
0C	Special function checksum error
OD	Special function invalid location address

NOTES

APPENDIX C
PROGRAMMING MODULE GENERAL AND ERROR MESSAGES

During programming module operation, descriptive messages may appear in the lower left part of the screen. These messages are of two types: general messages and error messages. General messages indicate that a programming module operation is proceeding normally, while error messages indicate that something is wrong — either an operator mistake or a system problem. Below is an alphabetical list of possible messages, the type of message (general or error), in what mode the message typically occurs, and an explanation of the message.

MESSAGE	TYPE	MODE	EXPLANATION
ADDR OUT OF BOUNDS	Error	All	There is no KIB rack entry for the signal address you have entered.
BAD CONTROLLER MESSAGE	Error	All	The programming module is unable to decipher a message from the sequencer. May indicate system problems.
BAD SIGNAL	Error	Build String	The string you are trying" to test has a signal that does not reference an existing KIB rack entry.
COMMAND COMPLETE	General	All	The string under test has been entered into sequencer memory.
CONTROLLER MEMORY EXCEEDED	Error	Build String	The available application software memory is full. No more strings can be entered into sequencer memory.
CURRENT GROUP IS > 2 FOR HIOB	Error	Supvr.	There can be only 2 groups assigned to a particular HIOB station.
DUPLICATE COIL	Error	Build String	A string with this coil signal address already exists in sequencer memory.
DUPLICATE RACK DEFINITION	Error	Supvr.	You have attempted to create a rack entry that duplicates a rack entry already present.
END OF RACK TABLE	General	Supvr.	The KIB rack entry displayed is the last one in the system.

MESSAGE	TYPE	-MODE	EXPLANATION
EXEC 00	Error	All	System problem - contact General Electric Company.
EXEC 01	Error	All	System problem - contact General Electric Company.
FORWARD REFERENCE TABLE FULL	Error	Build String	An internal sequencer table is full and no more strings can be entered into sequencer memory.
FROM ADDR IS > TO ADDR	Error	Supvr.	The "From" address you have entered must be smaller than the "To" address.
HMHA ADDRESS INVALID	Error	Supvr.	The HMHA address specified as a system parameter must be a multiple of hexadecimal 800.
I/O FAILURE - RETRYING	Error	Display String	Communication between the programming module and sequencer has been interrupted during string status update in Display String mode. The programming module is attempting to retransmit the status request.
I/O FAILURE - TRY AGAIN	Error	All	The programming module is unable to communicate with the sequencer. You should attempt the operation again.
ILLEGAL CONTROLLER MESSAGE	Error	All	The programming module is unable to decipher a message from the sequencer. May indicate system problems.
ILLEGAL ENTRY	Error	All	You have pressed a key or attempted a function which cannot be used in the current mode.
ILLEGAL STRING	Error	Build String	You are attempting to test a string whose string number or coil signal address already exists in another string.
INVALID COMMAND	Error	All	The sequencer is unable to decipher a message from the programming module.
IO FAILURE	Error	All	The programming module is unable to communicate with the sequencer or you may be trying to force a signal that does not exist. May indicate an unplugged signal cable, a sequencer without power, or a system problem.
KEY SWITCH NOT ENABLED	Error	All	You are attempting to modify sequencer memory with the MODIFY LOGIC keyswitch in the OFF position.

MESSAGE	TYPE	MODE	EXPLANATION
LIST FULL, CHANGE ADDR RANGE	General	Supvr.	The hardware diagnostic has filled the screen with data. To complete the test, change the address range and perform the diagnostic again.
NO ENTRIES IN RACK TABLE	General	Supvr.	No rack entries have been made in the KIB rack table.
NOT IN OFF LINE MODE	Error	supvr.	You are attempting to perform a KIB function without taking the sequencer off-line.
ON-LINE INITIALIZATION FAILURE	Error	Supvr.	This failure occurs while trying to bring the sequencer on-line. It can be caused by a hardware initialization failure or incompatibilities between product software and hardware. If this error message appears, press BUILD STRING and note the self-test error code that appears on the screen. Appendix B will tell you what the self-test error code means.
OPERATION COMPLETE	General	All	The operation you requested has been completed.
OPERATION STOPPED • INCOMPLETE	Error	All	The current tape operation (read, write, or compare) has been stopped due to a sequencer problem.
PRESS SHIFT AND ENTER	General	All	You must press SHIFT and then ENTER to complete the operation.
PRINTER FAILURE	Error	S upvr.	The printer is not responding to commands from the programming module. May indicate an unplugged signal cable, a printer without power, or a system problem.
RACK DEFINITION TABLE OVERFLOW	Error	Supvr.	There cannot be more than 32 rack entries in the KIB table.
RECEIVE FAILURE	Error	All	The programming module is unable to receive a message that the sequencer is attemping to send.
SEARCHING FOR BLOCK	General	B u i 1 d or Display String	The sequencer is attempting to retrieve a string from sequencer memory. This message should appear only momentarily.
SIGNAL INVALID	Error	Display Signal	The signal you have requested to be displayed does not have a corresponding KIB rack entry.

MESSAGE	TYPE	MODE	EXPLANATION
SIGNAL NOT A POINT	Error	Display Signal	You are trying to force a signal that has not been specified as a point in the KIB rack entry table.
SIGNAL NOT DISPLAYED	Error	Display Signal	You are attempting to force a signal that is not currently displayed.
SIGNAL NOT DISPLAYED OR FORCED	Error	Display Signal	You are trying to change the state of a signal that is not currently displayed or one that is not under force control.
STRING ENTERED	General	Build String	A string has been checked and placed in an on-line test condition.
STRING NOT IN TEST	Error	Build String	You have attempted to enter a string into sequencer memory or remove a string from test state while that string is not under test.
SWEEP TIME NOT A MULTIPLE OF 10 MSEC	Error	Supvr.	The sweep time parameter must be a multiple of 10 milliseconds.
SYSTEM BUSY ▪ TRY AGAIN	Error	All	The programming module was busy when you made a request. Repeat the request.
TAPE FAILURE	Error	Supvr.	Programming module is unable to properly read from or write to the cassette tape. May indicate no data received from the tape drive, the data received from the tape drive is invalid, the tape is write protected, or no tape has been inserted into the tape drive.
TRANSMIT FAILURE	Error	All	The programming module is unable to send a message to the sequencer.
TRYING TO DELETE RACK WITH REFERENCES	Error	Supvr.	You are attempting to delete a KIB rack entry that is referenced by strings in sequencer memory.
TRYING TO MODIFY MAX GROUP	Error	Supvr.	You are attempting to modify the maximum group value in a KIB rack entry. The only way to do this is to delete all signals that reference the rack entry, delete the rack entry, then recreate the rack entry with a new maximum group value.
UNDEFINED SIGNAL	Error	Display Signal	A rack entry does not exist for the signal you wish displayed.

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